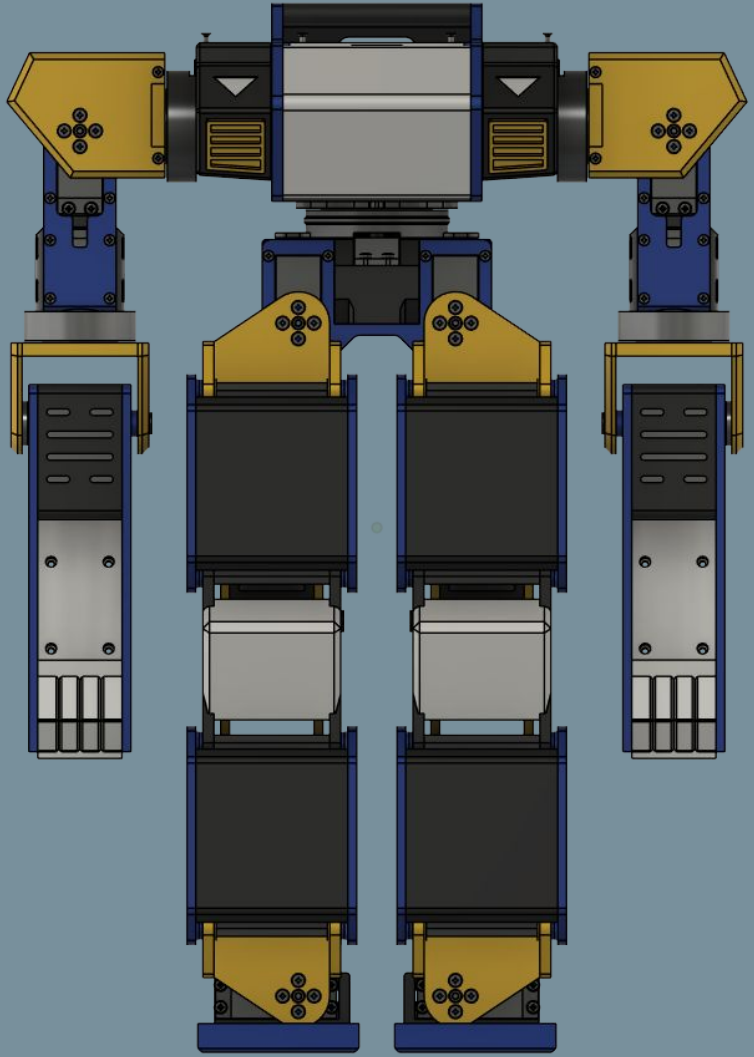
Endeavor Humanoid Robot Assembly Instructions

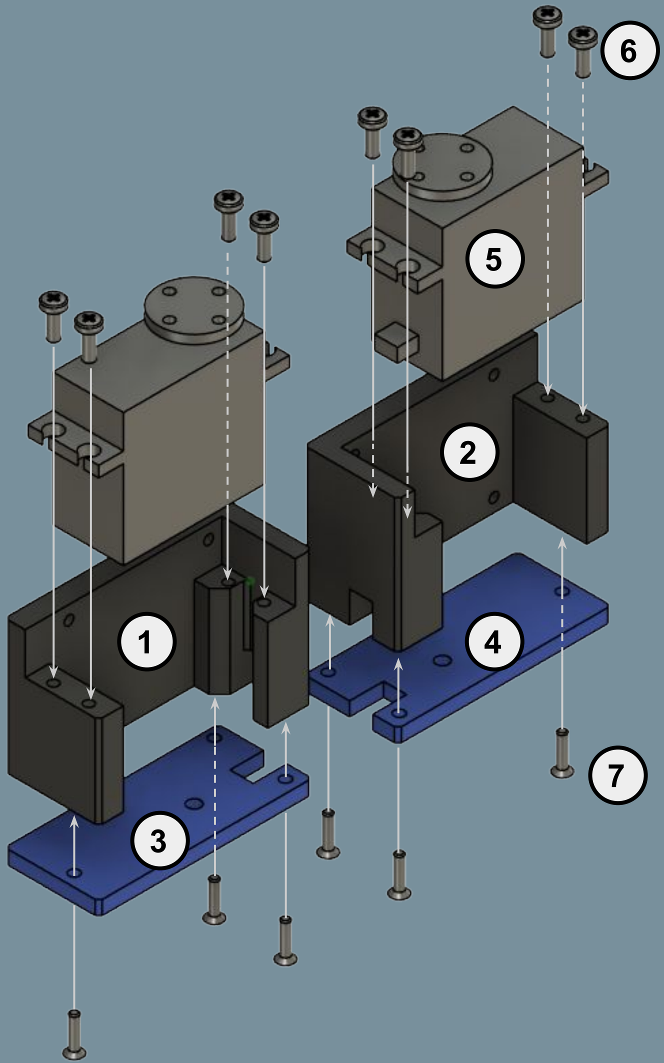
ProtoDane Robotics



Document Version: 1.0
Last Revised: Feb. 20, 2026

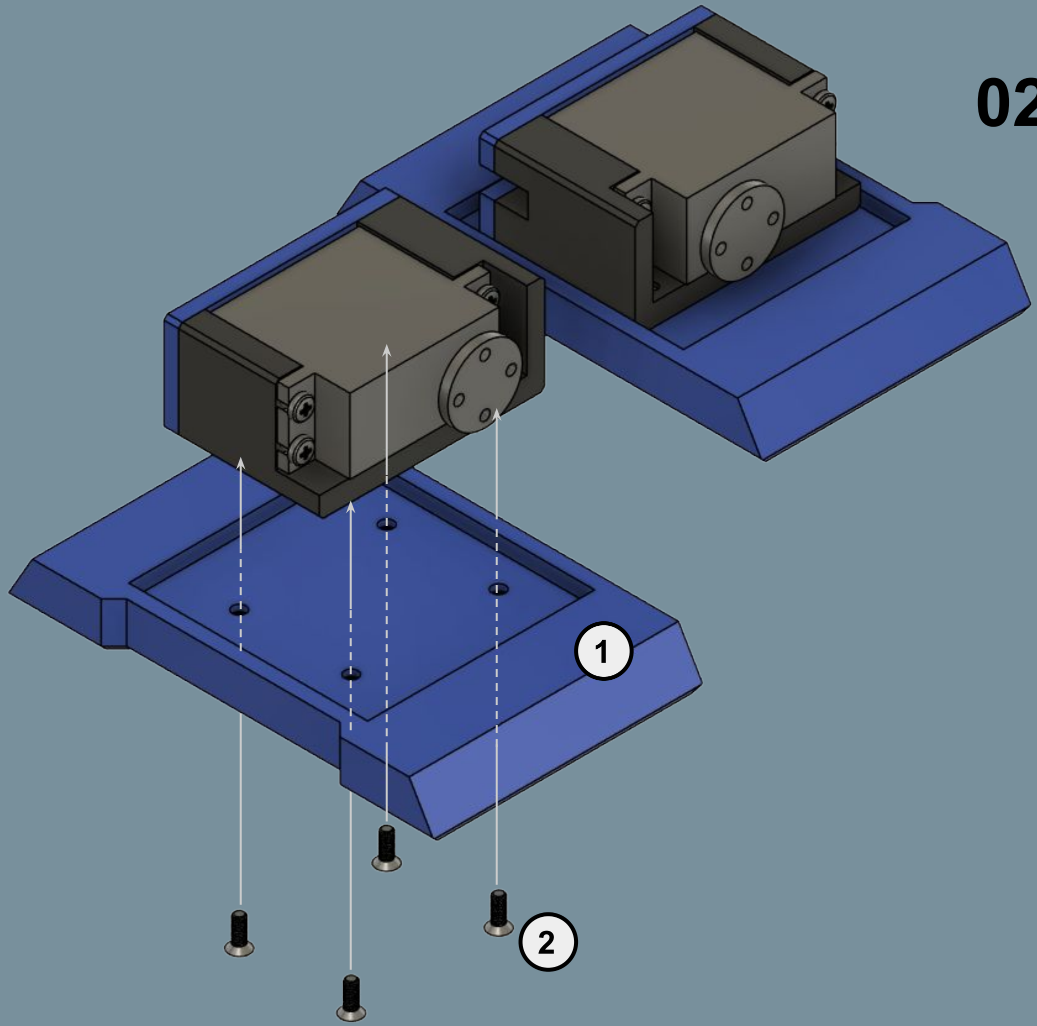
Notes

- Refer to servo calibration instructions for DS3235 servos. All servos should be calibrated before starting assembly. **Do not remove or rotate the servo disc to ensure servo joints are set properly when powered.**
- This assembly does not contain heat inserts. As such, all fasteners should be tightened properly, either by hand or motorized tool. **It is up to you to ensure that all fasteners are secured and not overtightened that it destroys the printed hole.**
- Read each step carefully. You are responsible for proper assembly, safe tool handling, and any risks associated with this project. If you have any questions, please reach out on the ProtoDane Discord server.



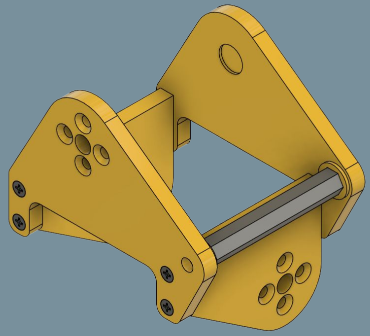
01: Ankle Servo Assembly

No.	Part Name	Qty
1	LeftAnkleServoHousing	1
2	RightAnkleServoHousing	1
3	LeftAnkleBracket	1
4	RightAnkleBracket	1
5	DS3235 Servo	2
6	M3x8 Pan Head Screw	8
7	M2.5x10 Flat Head Screw	6



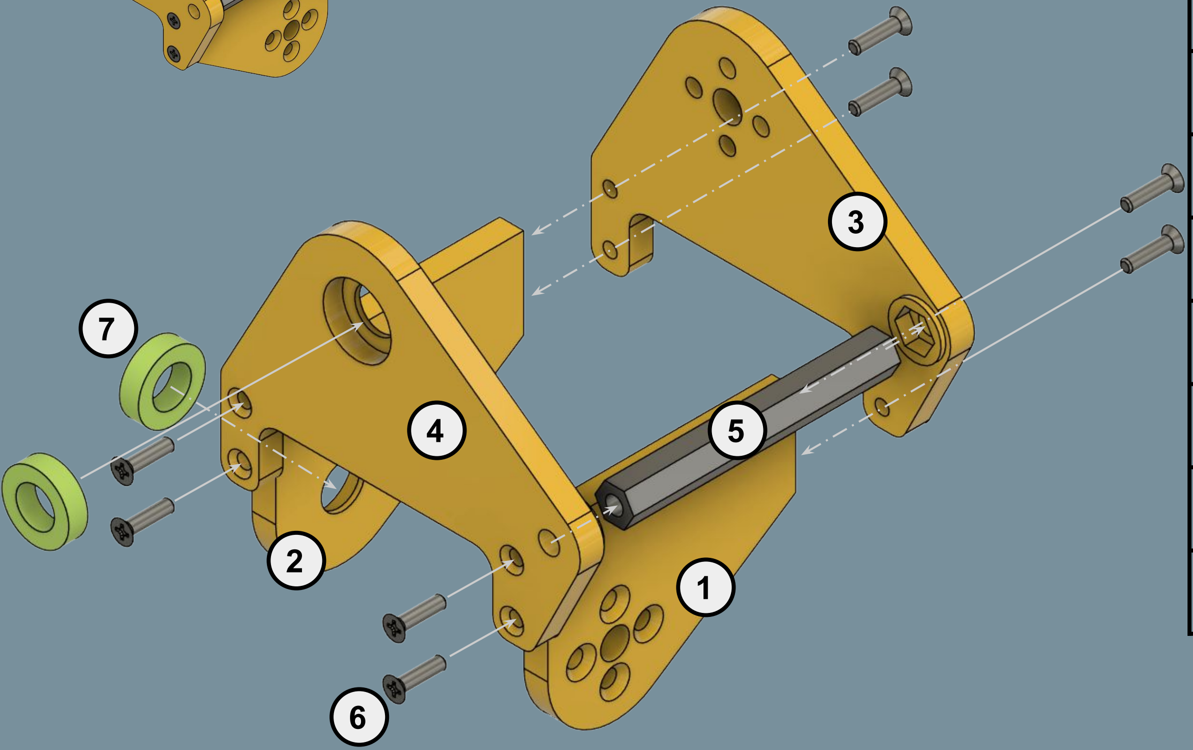
02: Foot Mounting

No.	Part Name	Qty
1	Foot	2
2	M3x8 Flat Head Screw	8



*Repeat step w/ mirrored parts

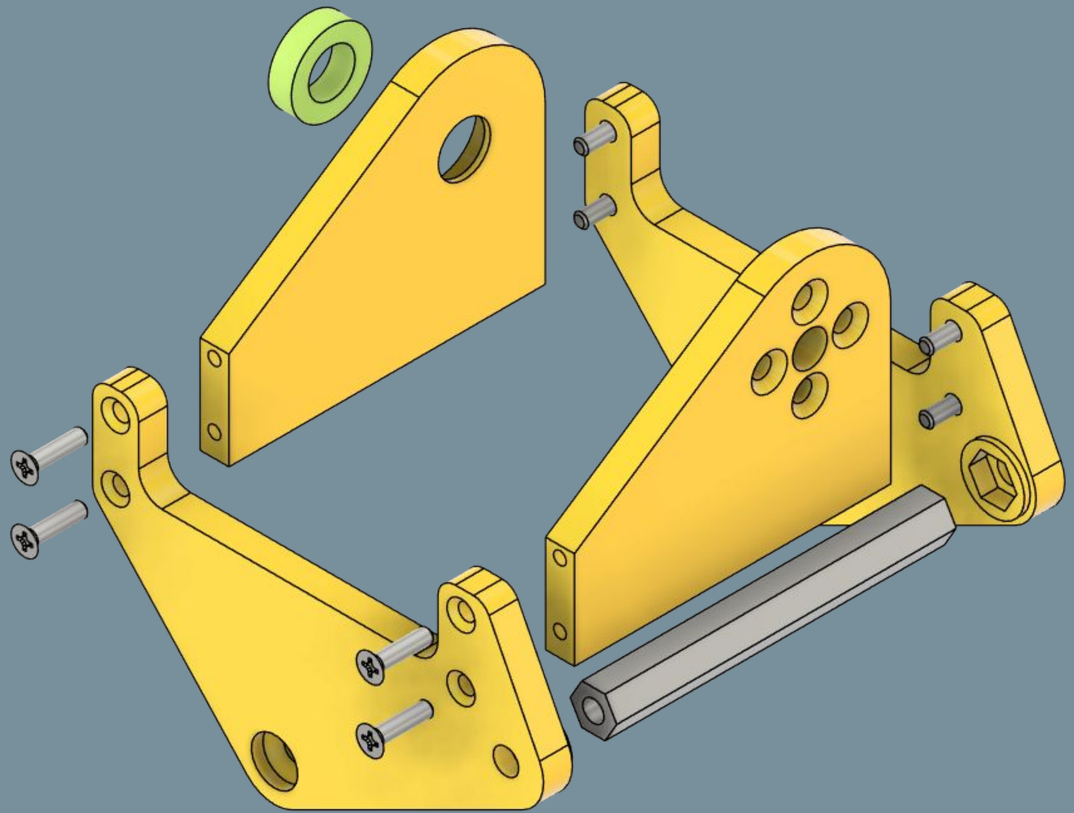
03: Ankle Bracket Assembly



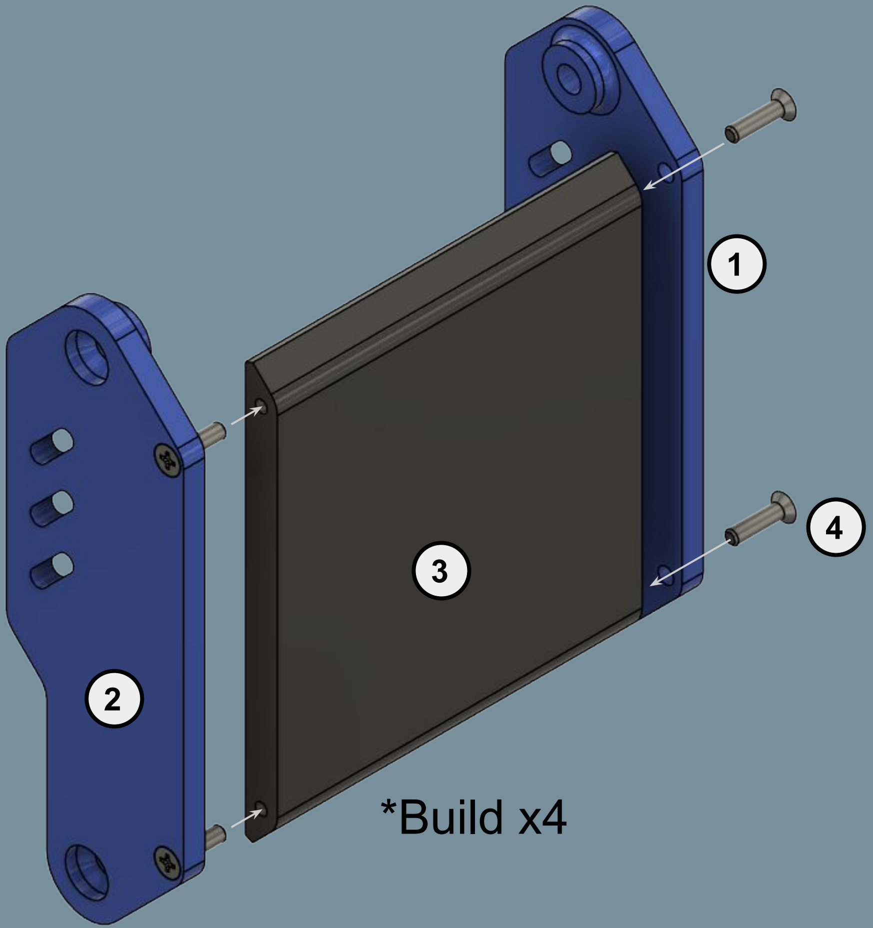
No.	Part Name	Qty
1	RollServoPlate (+Mirror)	1
2	RollBearingPlate (+Mirror)	1
3	AnkleServoPlate (+Mirror)	1
4	AnkleBearingPlate (+Mirror)	1
5	HexStandoffB	1
6	M2.5x10 Flat Head Screw	8
7	8ID 14OD 4T Bearing	2

*Repeat step w/ mirrored parts

04: Thigh Bracket Assembly

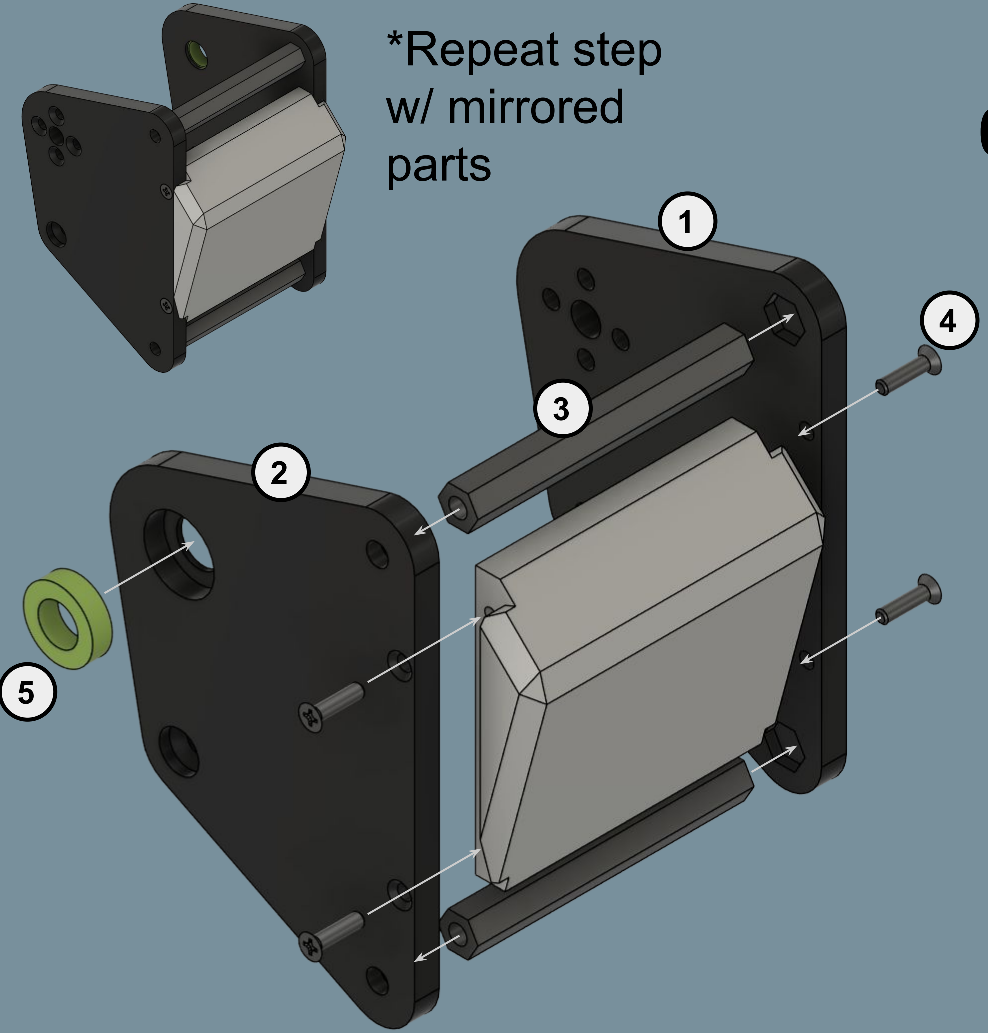


No.	Part Name	Qty
1	RollServoPlate (+Mirror)	1
2	RollBearingPlate (+Mirror)	1
3	ThighBracket	1
4	ThighBracket (Mirror)	1
5	HexStandoffB	1
6	M2.5x10 Flat Head Screw	8
7	8ID 14OD 4T Bearing	1



05: Front Leg Guards

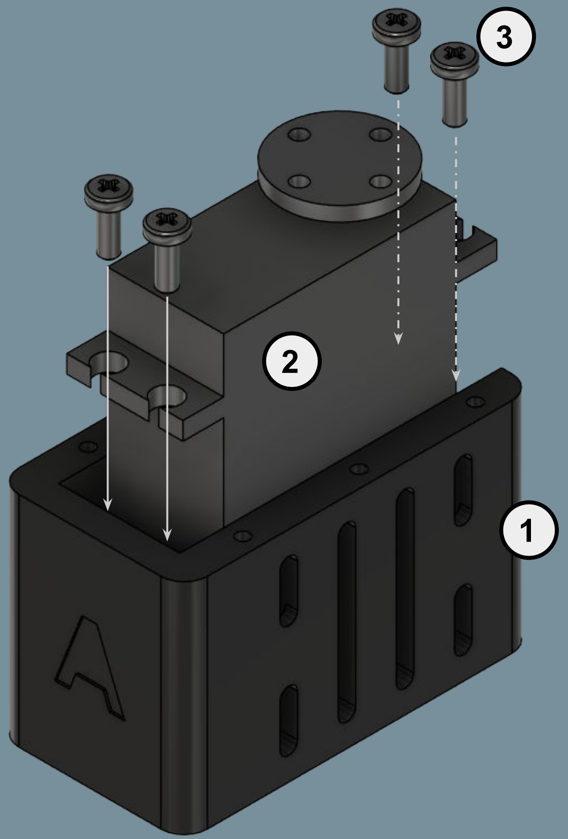
No.	Part Name	Qty
1	FrontLinkage	1(x4)
2	FrontLinkage(Mirror)	1(x4)
3	FrontLinkageCover	1(x4)
4	M2.5x10 Flat Head Screw	4(x4)



*Repeat step w/ mirrored parts

06: Knee Assembly

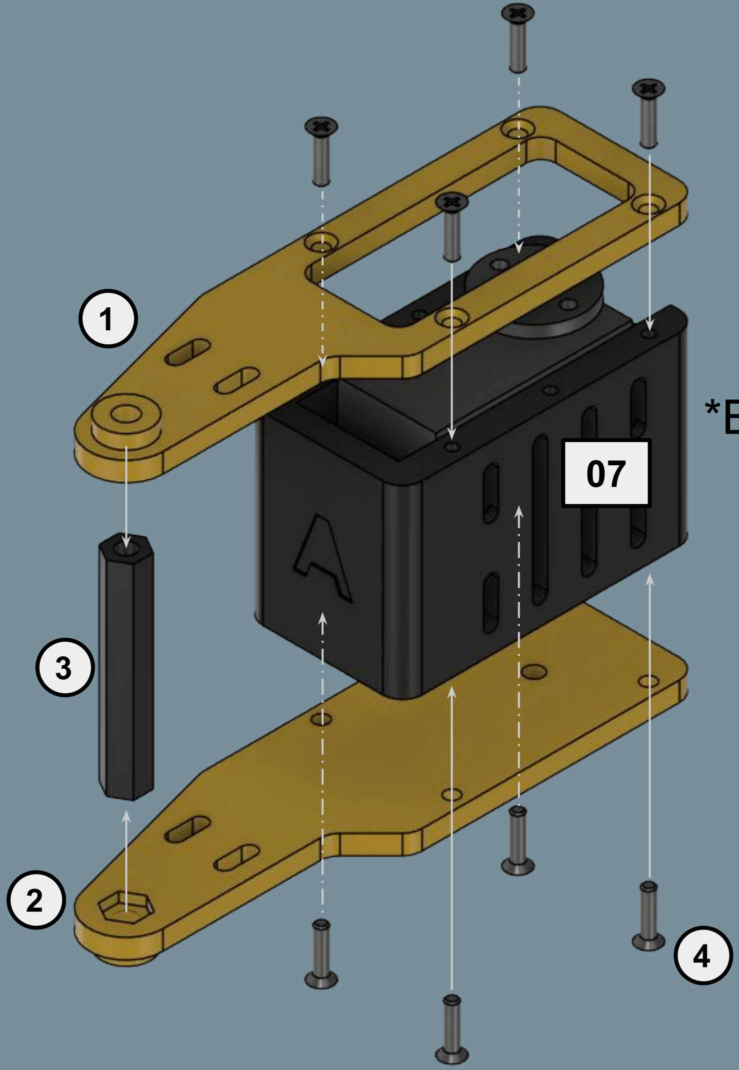
No.	Part Name	Qty
1	KneeServoPlate (+Mirror)	1
2	KneeBearingPlate (+Mirror)	1
3	HexStandoffB	2
4	M2.5x10 Flat Head Screw	4
5	8ID 14OD 4T Bearing	1



*Build x6

07: Servo Module A

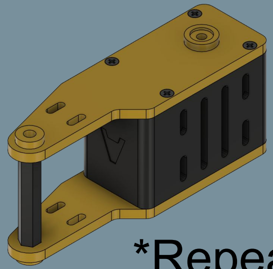
No.	Part Name	Qty
1	ServoHousingA	1x6
2	DS3235 Servo	1x6
3	M3x8 Pan Head Screw	4x6



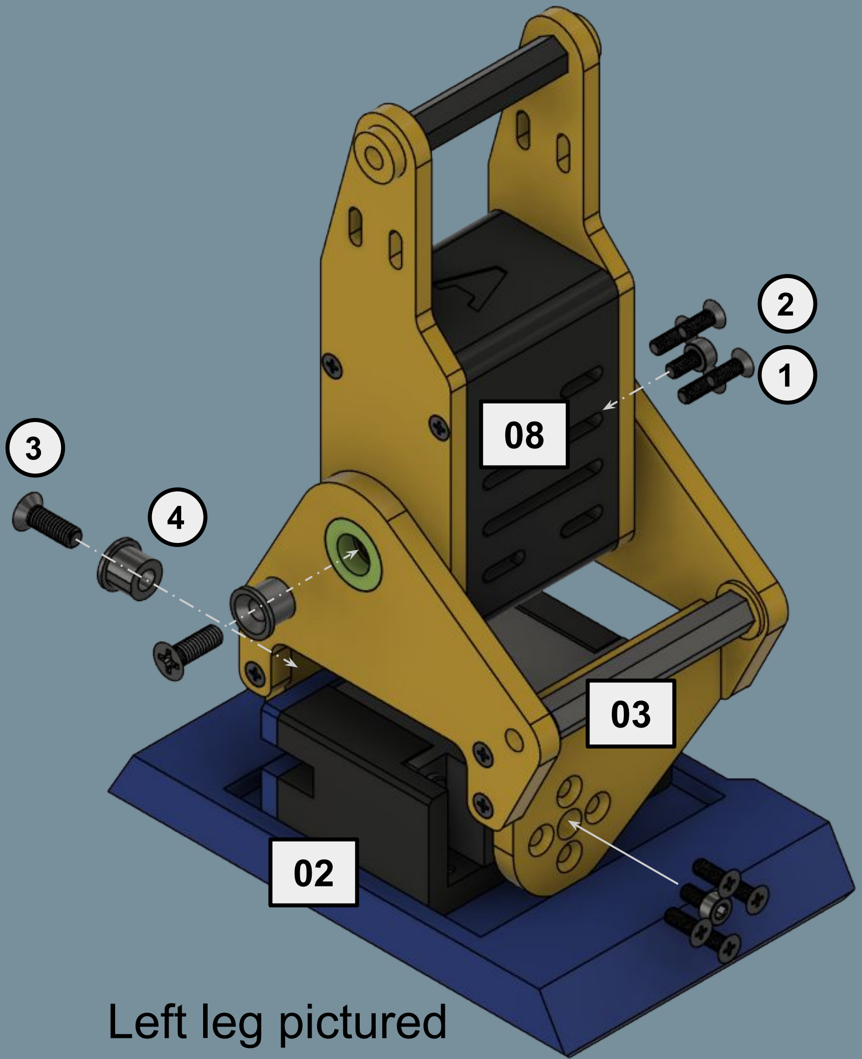
*Build x2

08: Servo Leg Assembly

No.	Part Name	Qty
1	LegServoBracket (+Mirror)	1x2
2	LegShaftBracket (+Mirror)	1x2
3	HexStandoffA	1x2
4	M2.5x10 Flat Head Screw	8x2



*Repeat step w/ mirrored parts



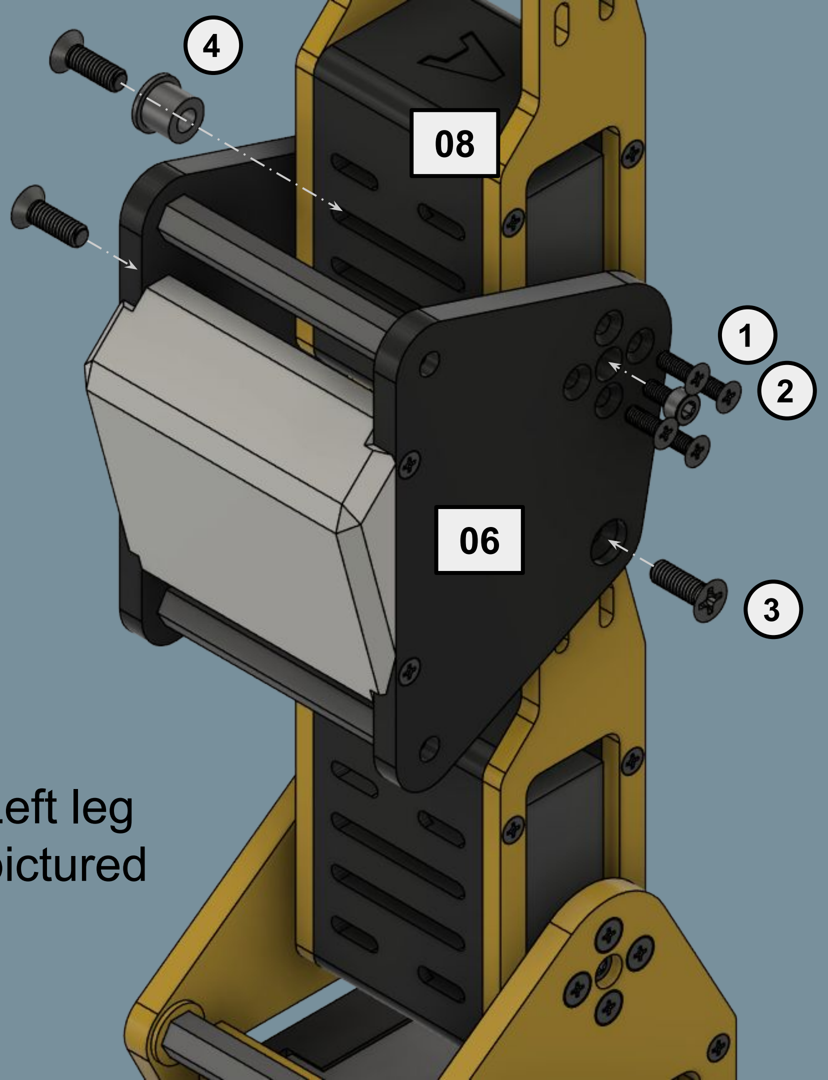
09a: Lower Leg Assembly

No.	Part Name	Qty
1	M3x8 Flat Head Screw	8
2	M3x6 Socket Head Screw	2
3	M4x12 Flat Head Screw	2
4	ShaftSleeve	2

Install in the orientation as shown; this is the servo's zero position

Left leg pictured

Repeat step for right leg



Left leg pictured

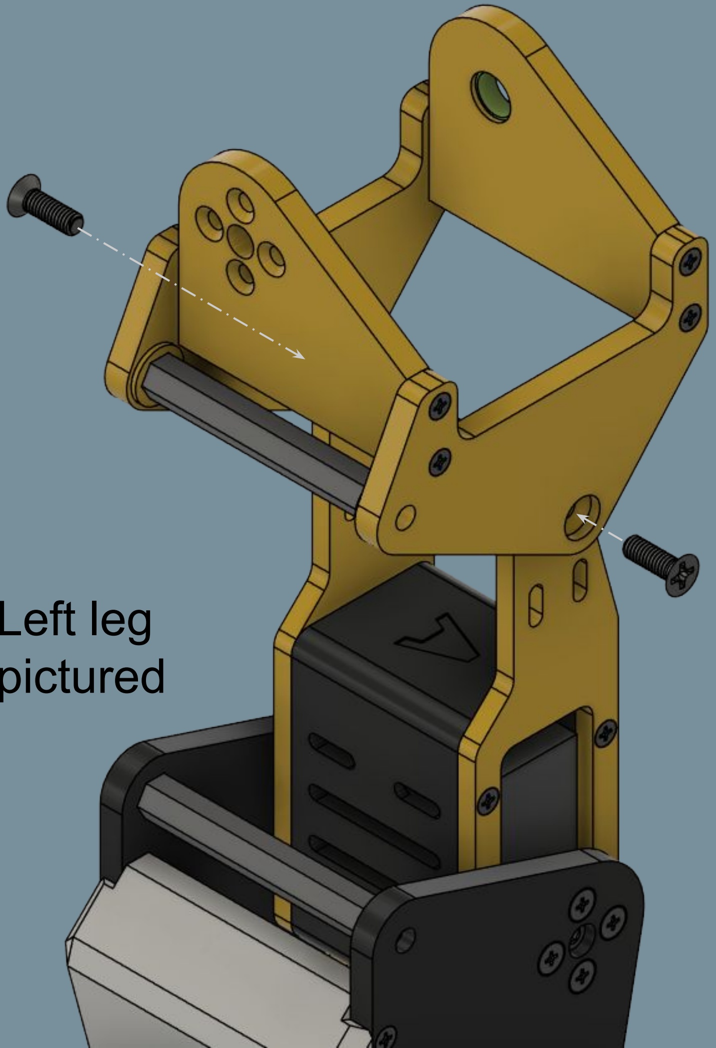
09b: Knee Mounting

No.	Part Name	Qty
1	M3x8 Flat Head Screw	4
2	M3x6 Socket Head Screw	1
3	M4x12 Flat Head Screw	3
4	ShaftSleeve	1

Install in the orientation as shown; this is the servo’s zero position

Repeat step for right leg

09c: Thigh Bracket Mounting



Left leg pictured

No.	Part Name	Qty
1	M4x12 Flat Head Screw	2

Repeat for right leg

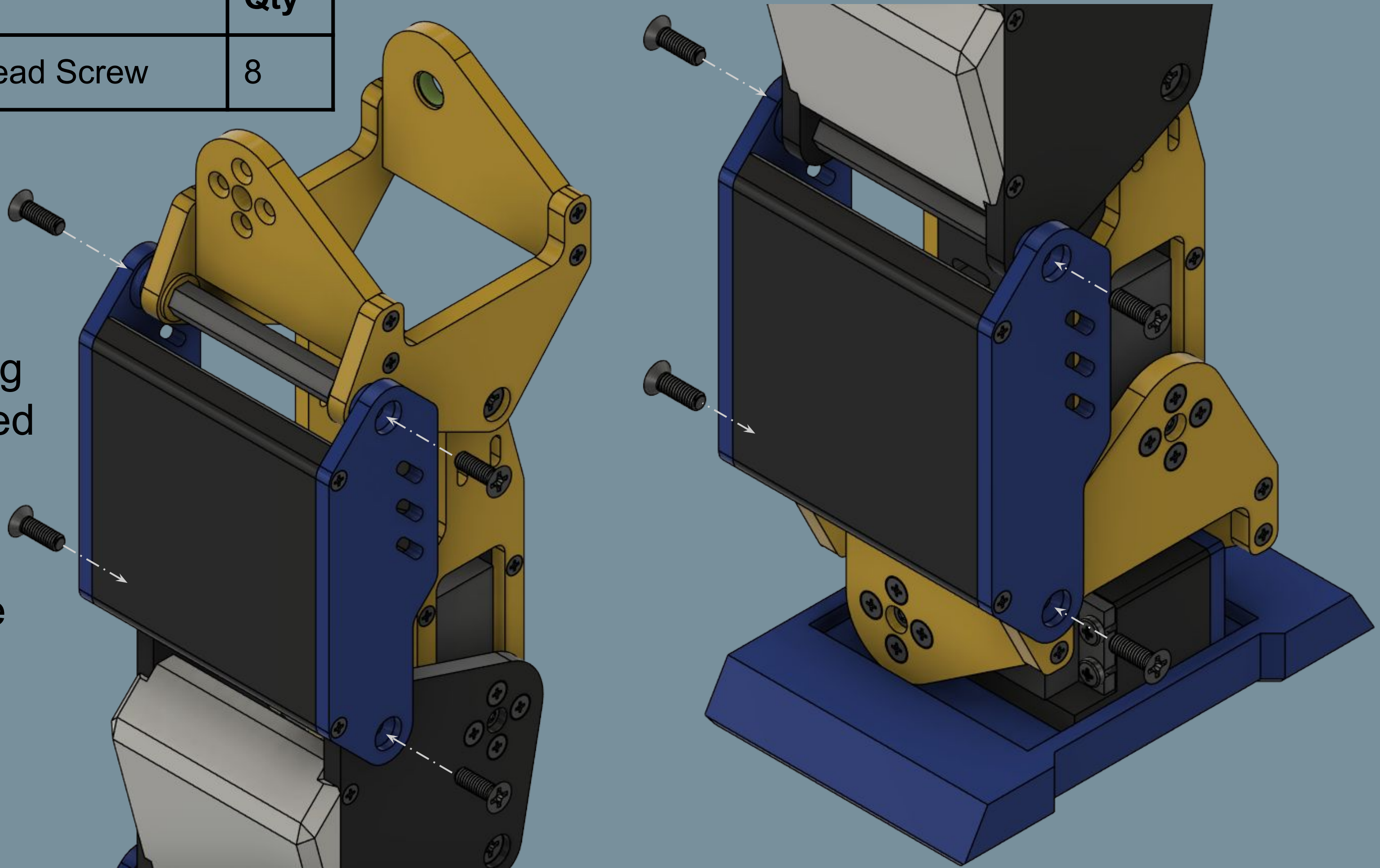
09d: Front Linkage Cover Mounting

No.	Part Name	Qty
1	M4x12 Flat Head Screw	8

Left leg pictured

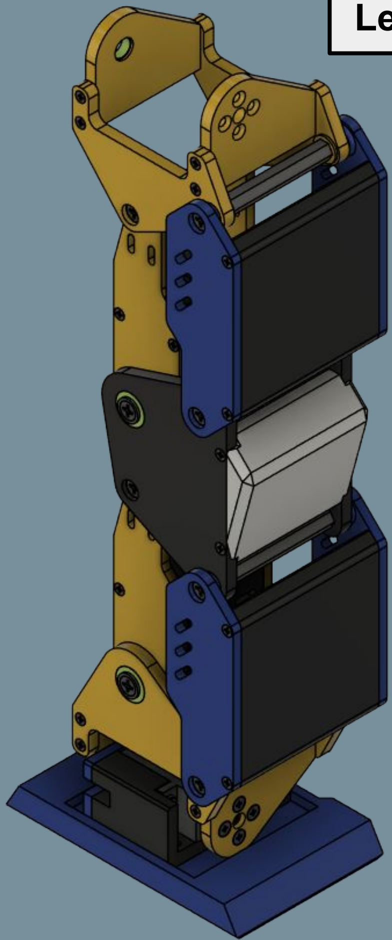
Tune screw tightness for free rotation

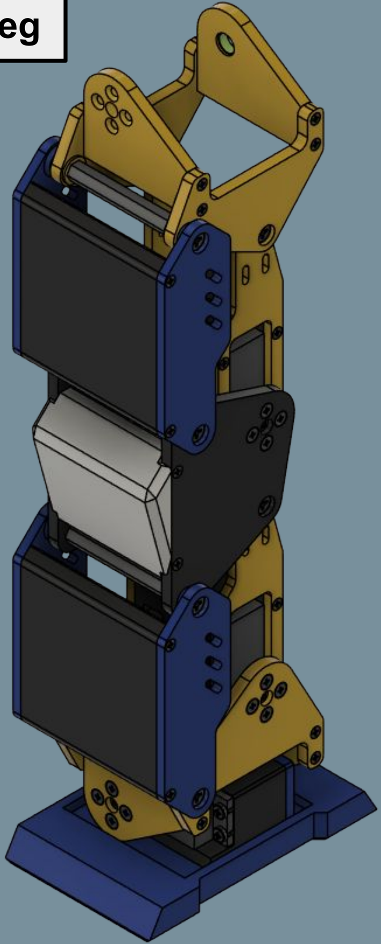
Repeat step for right leg



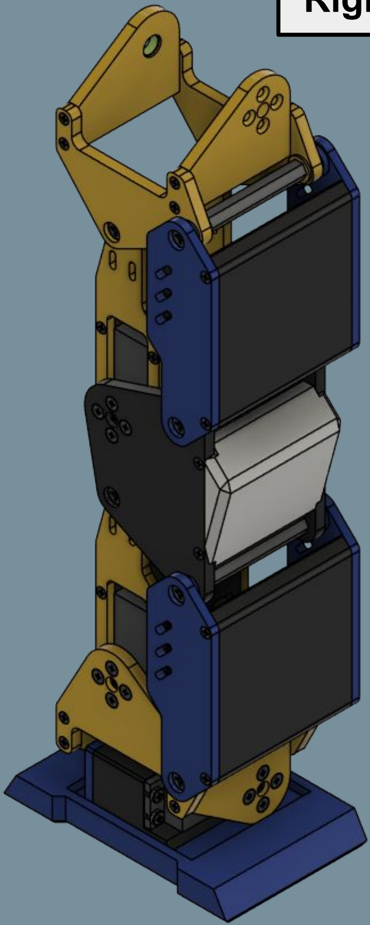
09e: Full Leg

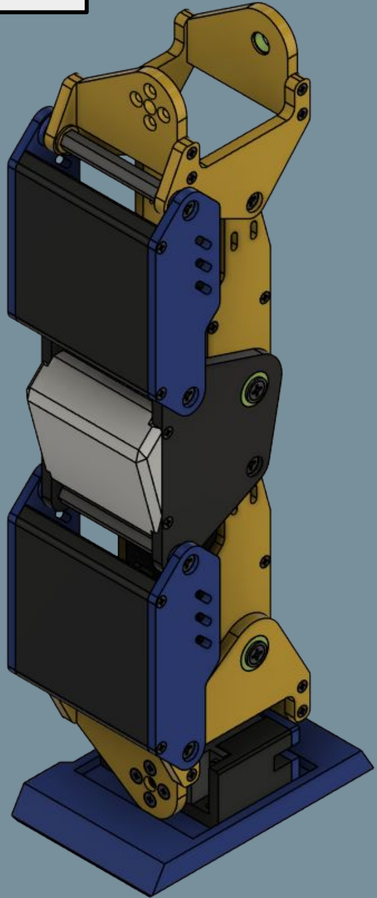
Left Leg





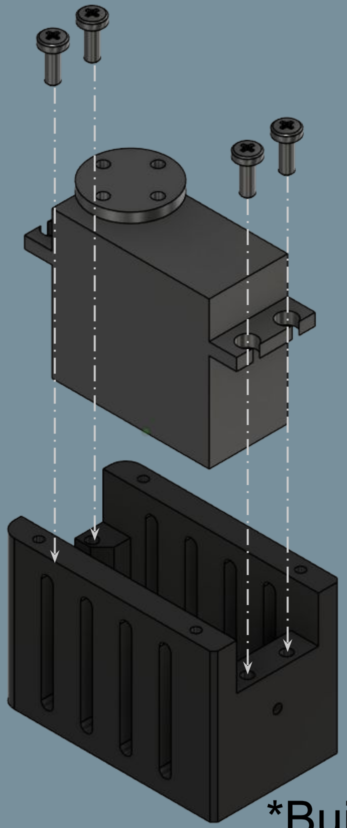
Right Leg



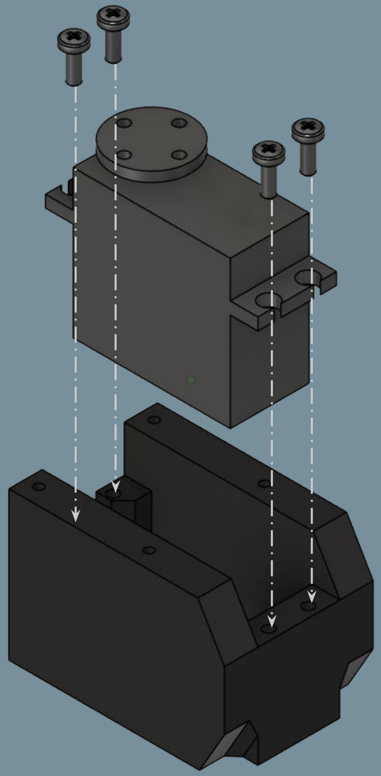


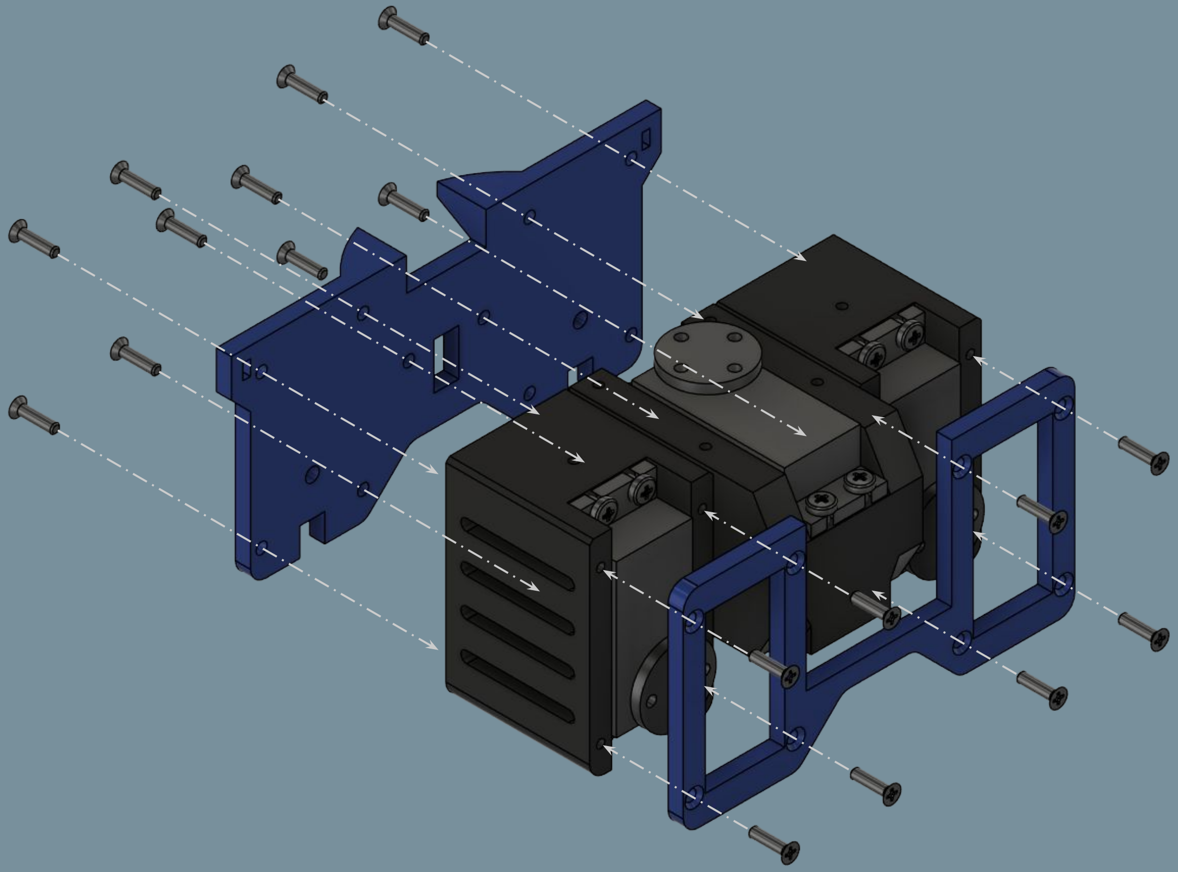
10: Pelvis Servo Modules

No.	Part Name	Qty
1	ThighServoHousing	2
2	TorsoServoHousing	1
3	DS3235 Servo	3
4	M3x8 Pan Head Screw	12



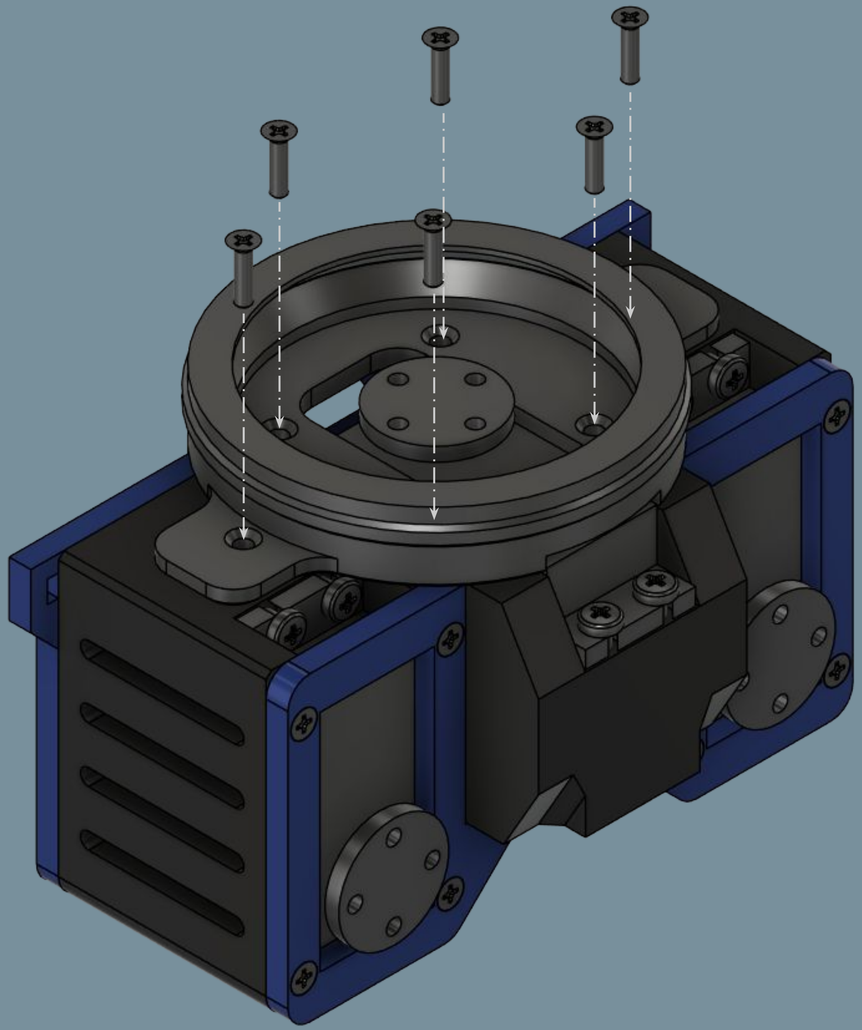
*Build x2





11: Pelvis Assembly

No.	Part Name	Qty
1	PelvisFrontBracket	1
2	PelvisRearBracket	1
3	M2.5x10 Flat Head Screw	18

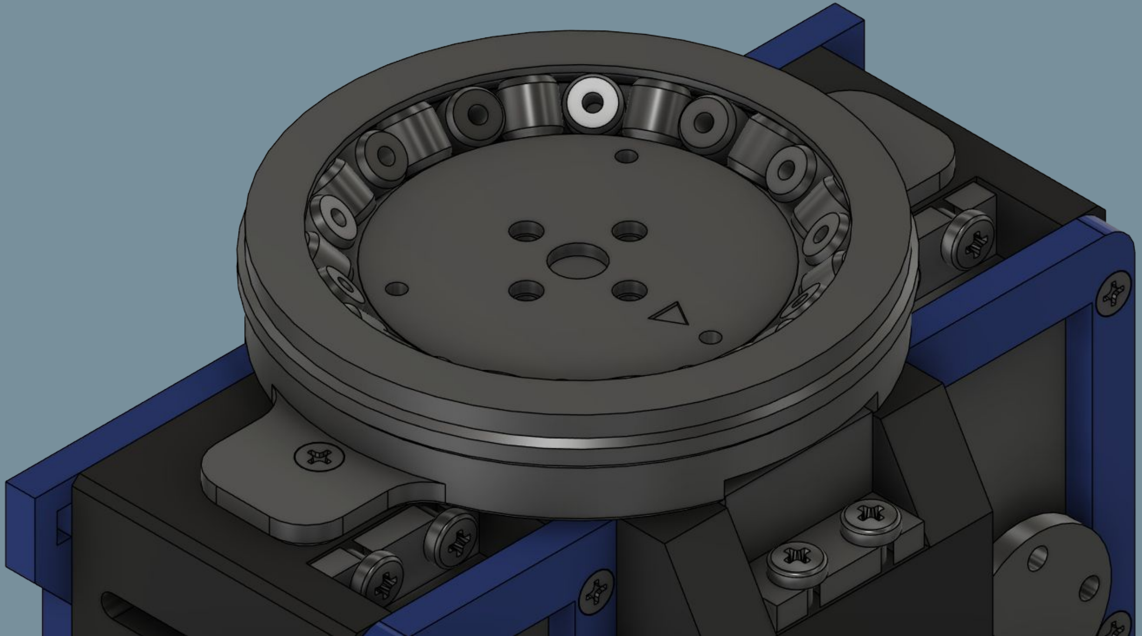


12a: Torso Slew Bearing Mounting

No.	Part Name	Qty
1	TorsoSlew-OuterRace	1
2	M2.5x10 Flat Head Screw	6

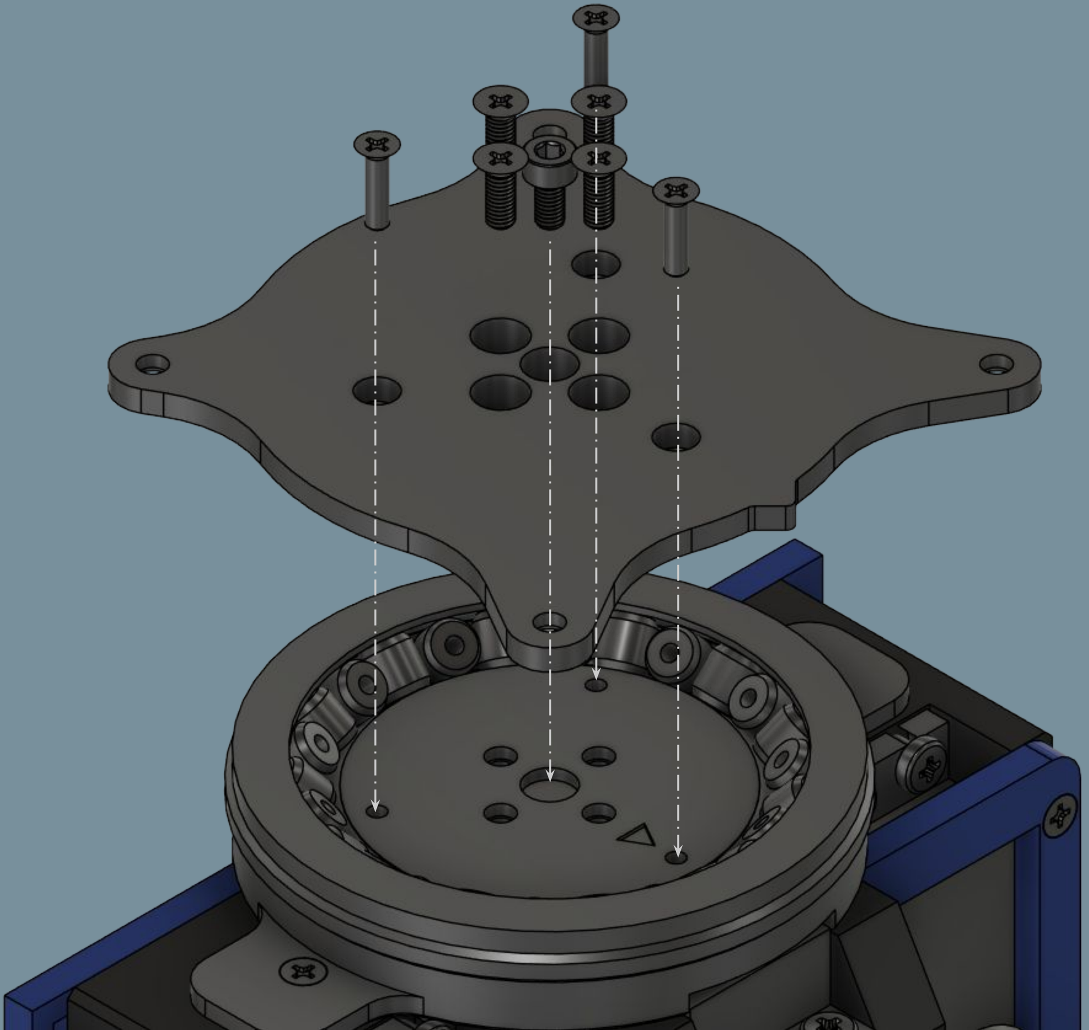
Drop the InnerBottom piece into the bearing housing first.

Alternate slew rollers by 90 degrees (see image).



12b: Lower Cap + Rollers

No.	Part Name	Qty
1	TorsoSlew-InnerBottom	1
2	SlewRoller	26

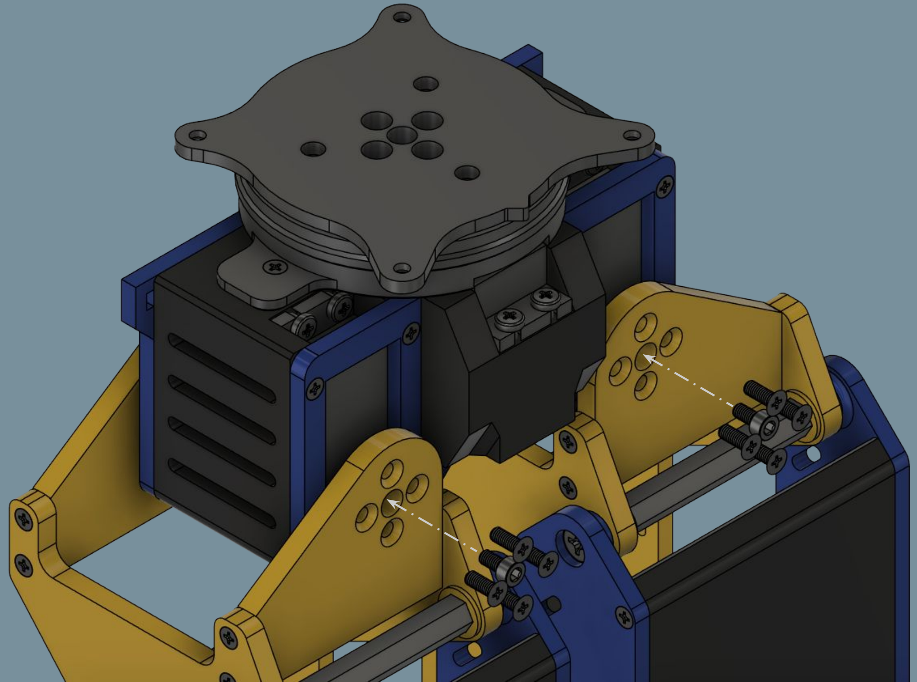


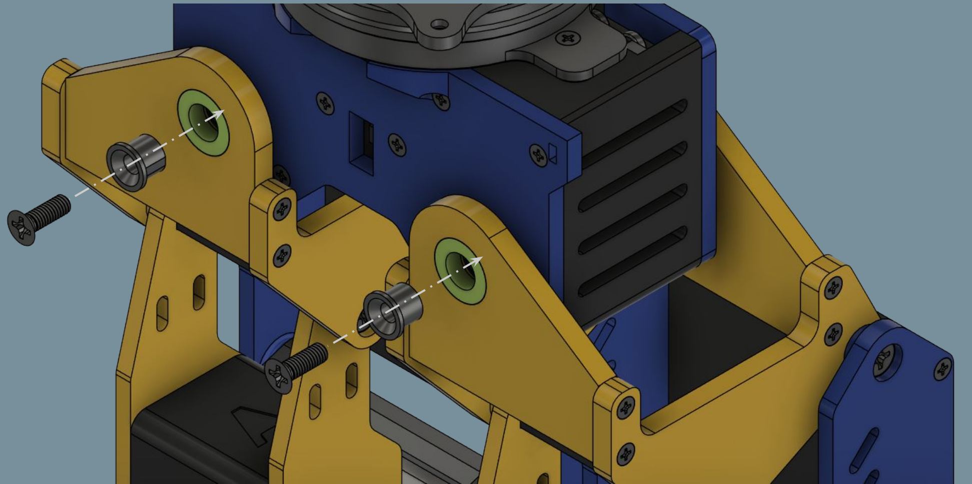
12c: Upper Cap Mounting

No.	Part Name	Qty
1	TorsoSlew-InnerTop	2
2	M2.5x10 Flat Head Screw	3
3	M3x8 Flat Head Screw	4
4	M3x6 Socket Head Screw	1

13: Pelvis/Leg Mounting

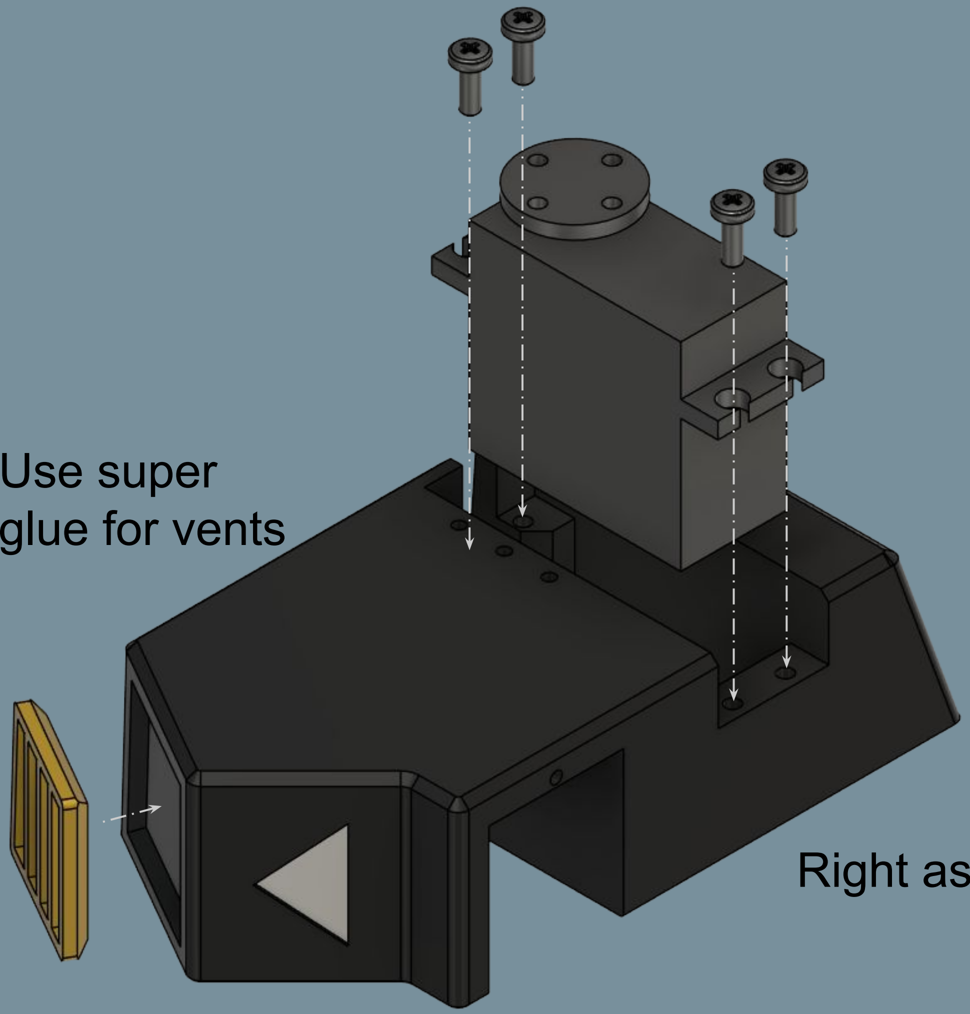
No.	Part Name	Qty
1	M3x8 Flat Head Screw	8
2	M3x6 Socket Head Screw	2
3	M4x12 Flat Head Screw	2
4	ShaftSleeve	2





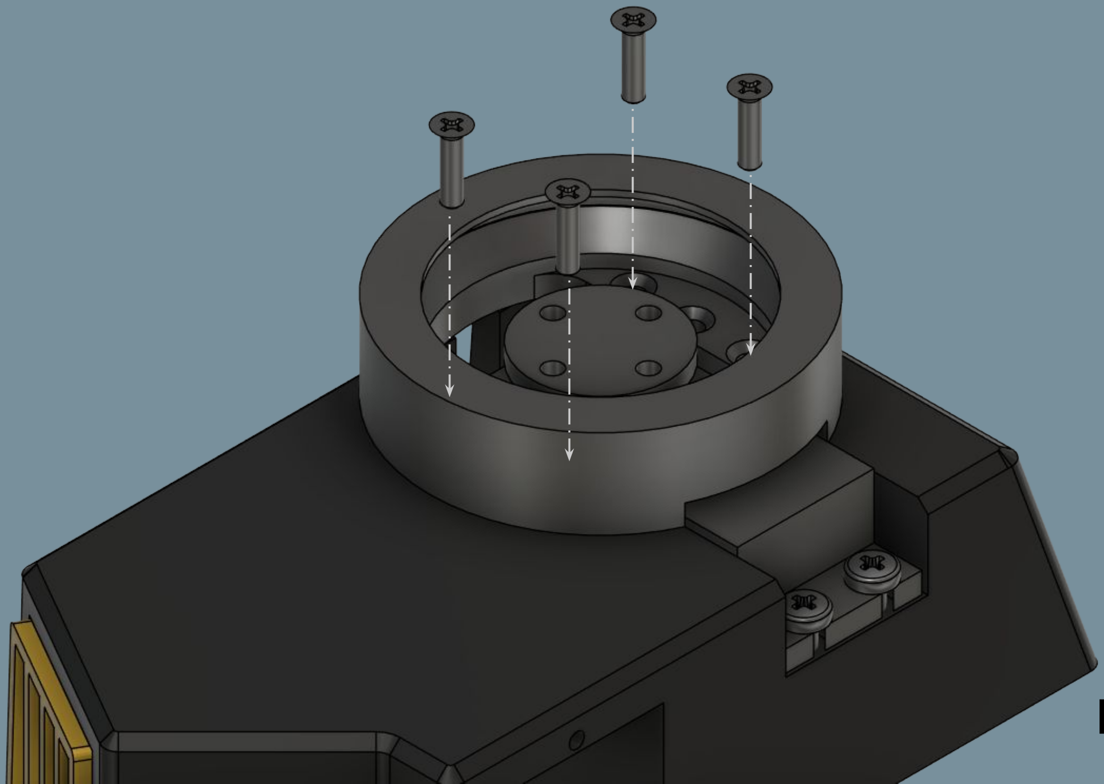
14a: Chest Servo Housing

No.	Part Name	Qty
1	ChestHousingL/R	1
2	Left/RightVent	1
3	DS3235 Servo (270deg)	1
4	M3x8 Pan Head Screw	4



Use super glue for vents

Right assembly pictured



14b: Arm Slew Bearing Mounting

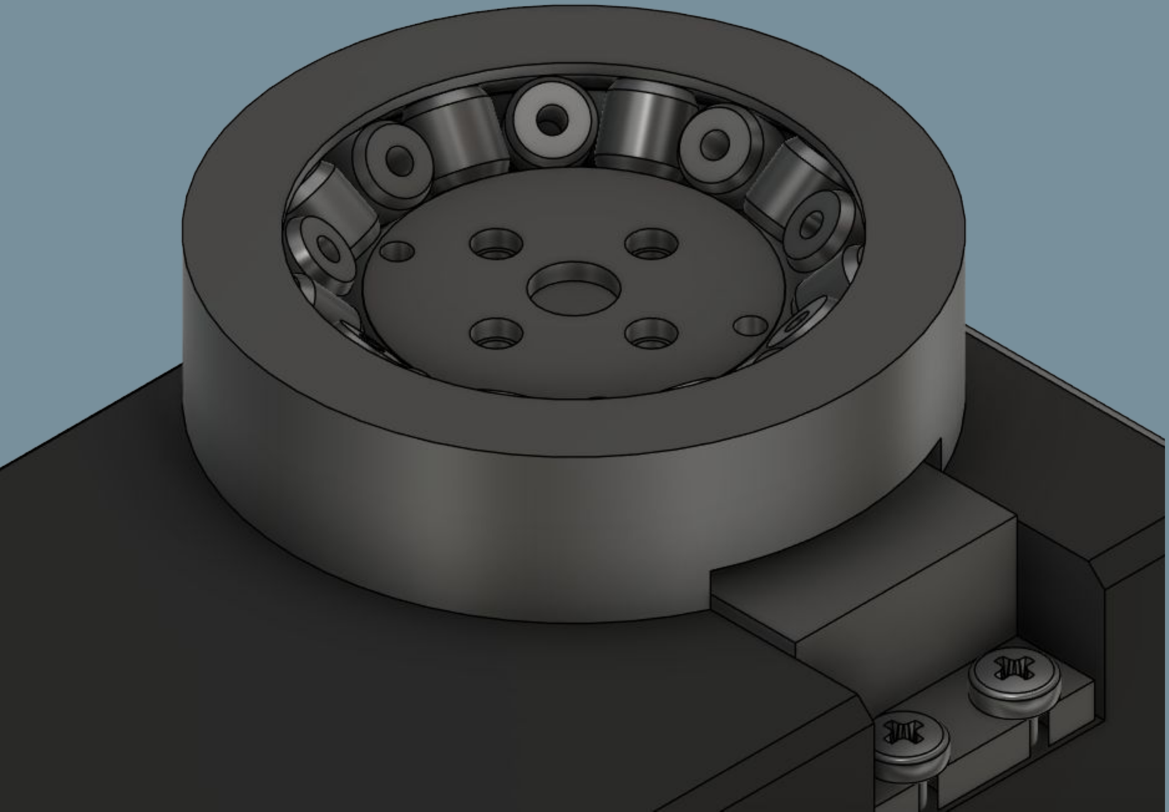
No.	Part Name	Qty
1	ArmSlew-OuterRace	1
2	M2.5x10 Flat Head Screw	4*

**3x screws okay if arranged in triangular pattern*

Right assembly pictured

Drop the InnerBottom piece into the bearing housing first.

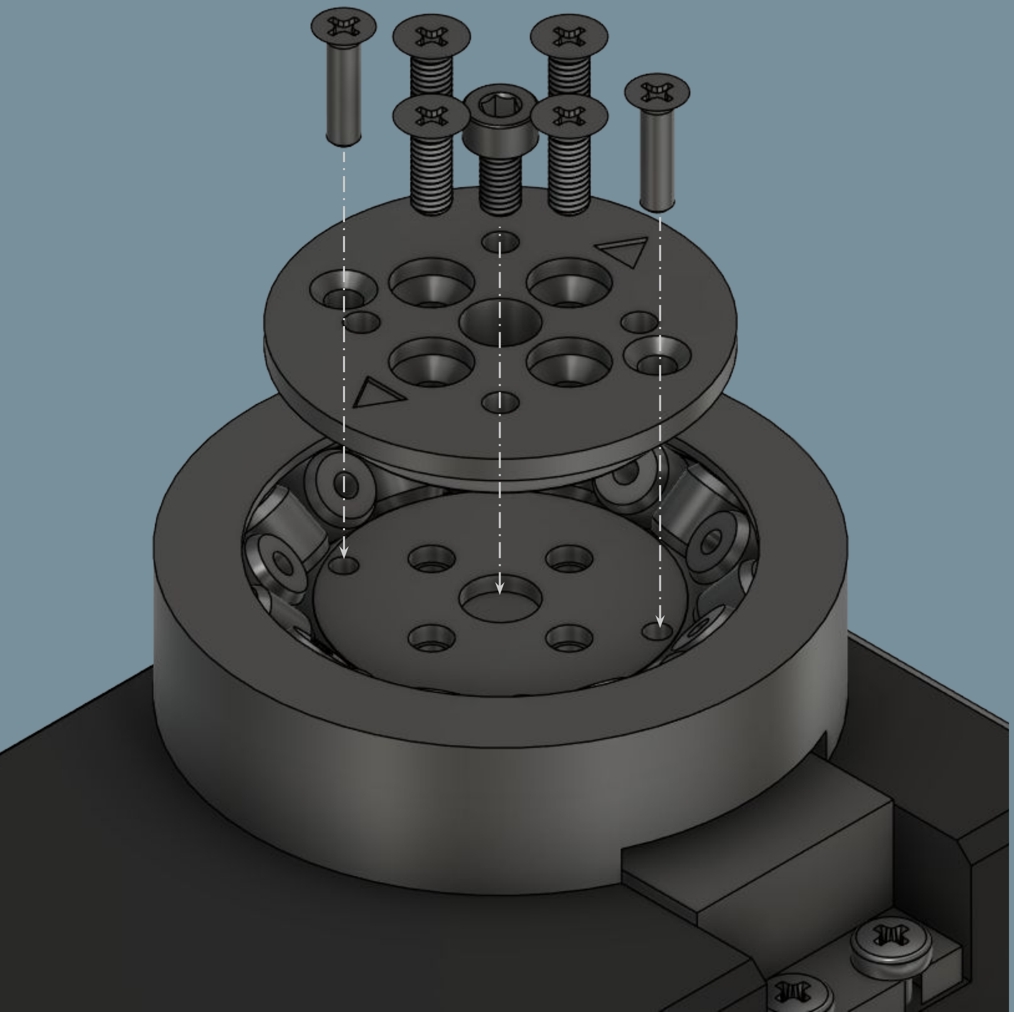
Alternate slew rollers by 90 degrees (see image).



14c: Lower Cap + Rollers

No.	Part Name	Qty
1	ArmSlew-InnerBottom	1
2	SlewRoller	18

Right assembly pictured



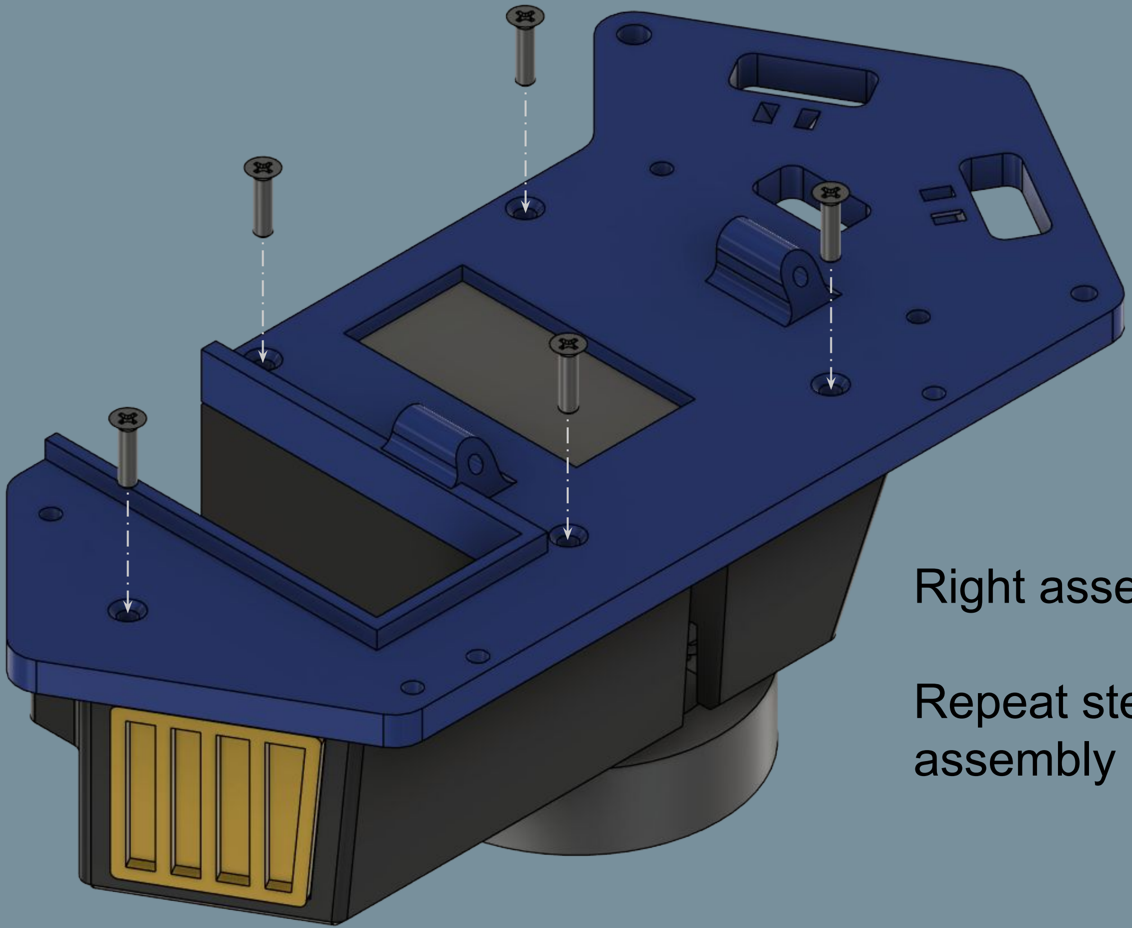
14d: Upper Cap

No.	Part Name	Qty
1	ArmSlew-InnerTop	1
2	M2.5x10 Flat Head Screw	2
3	M3x8 Flat Head Screw	4
4	M3x6 Socket Head Screw	1

Right assembly pictured

14e: Side Plate Mounting

No.	Part Name	Qty
1	TorsoLeft/RightPlate	1
2	M2.5x10 Flat Head Screw	5



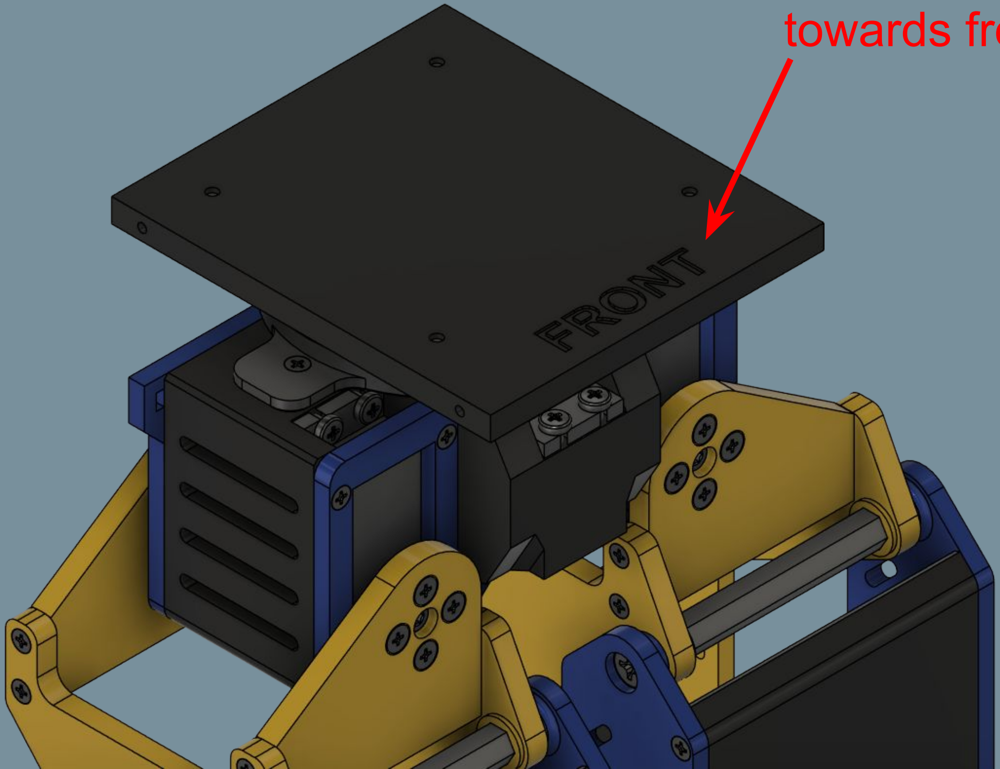
Right assembly pictured

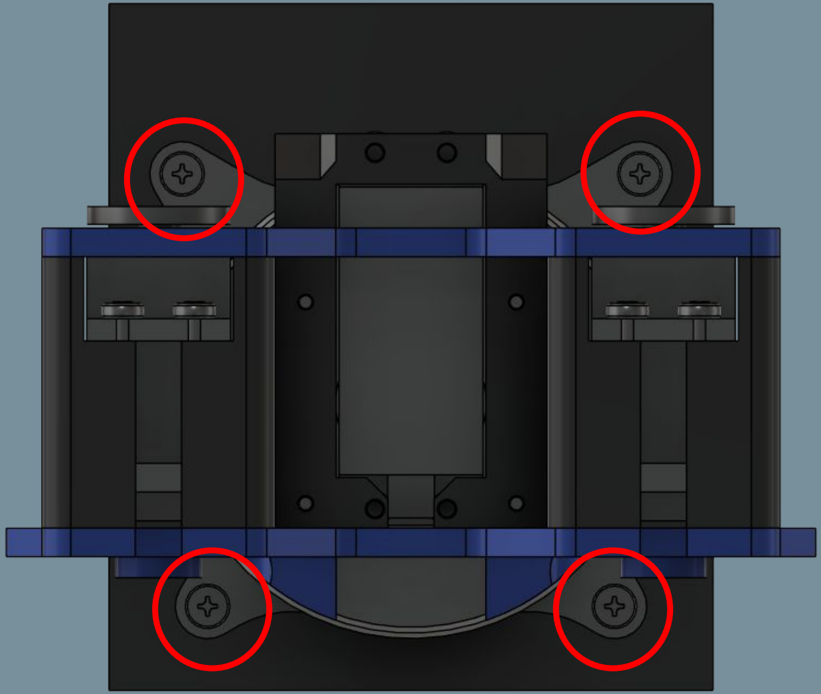
Repeat steps 14a-14e for left assembly

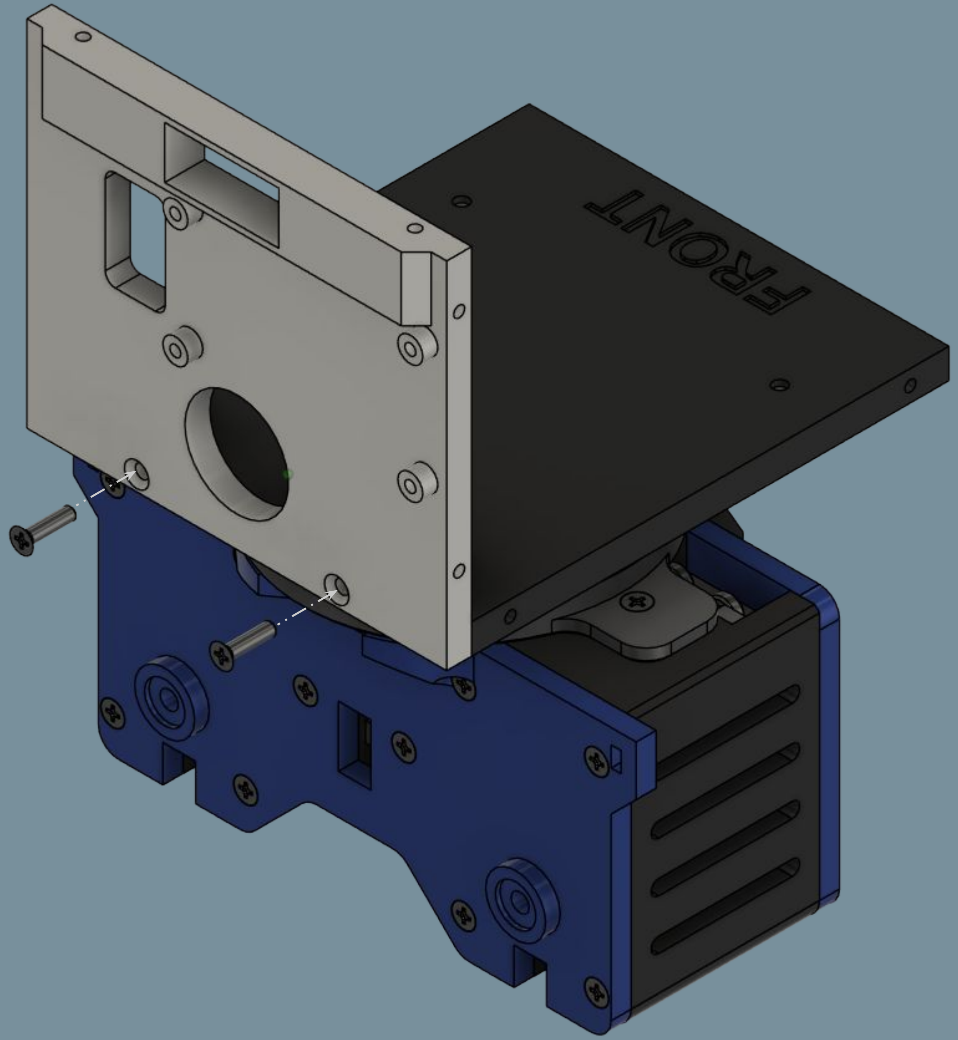
No.	Part Name	Qty
1	TorsoBottomPlate	1
2	M3x8 Flat Head Screw	4

15a: Torso Bottom Plate Mounting

Front text facing up and towards front of robot





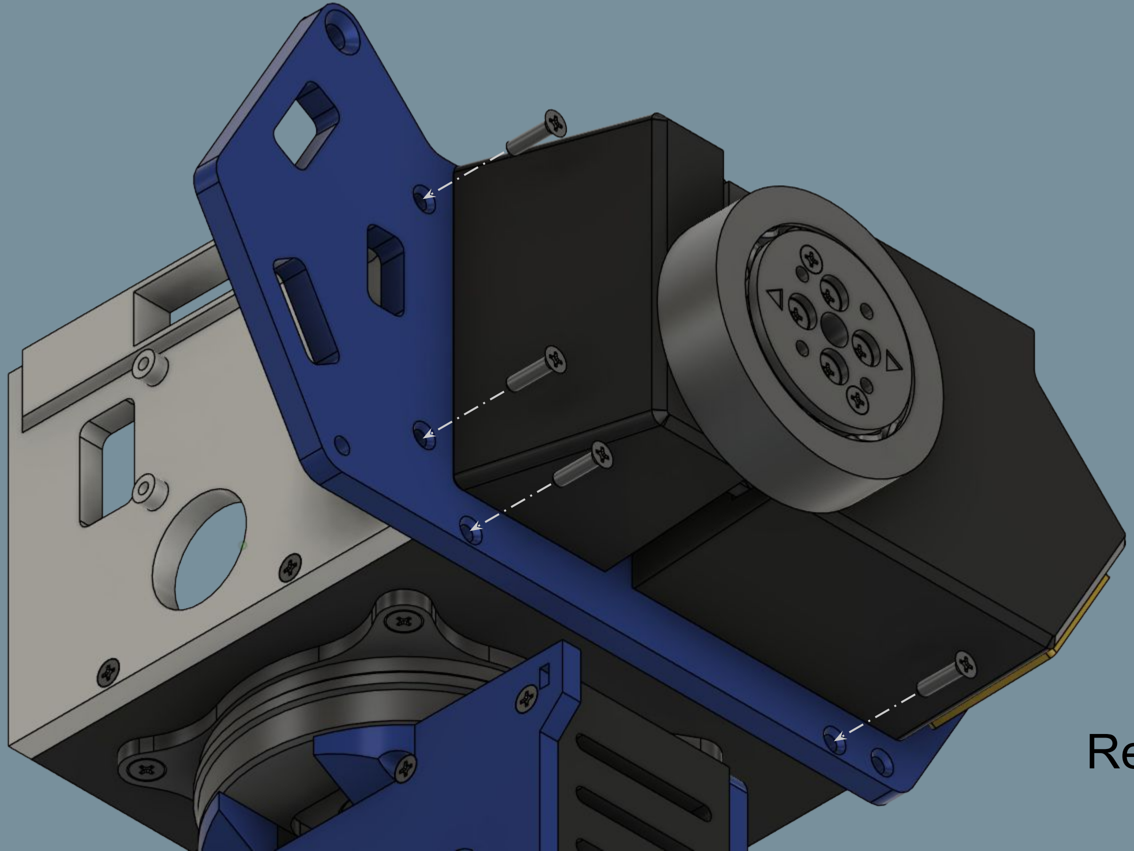


15b: Back Plate Mounting

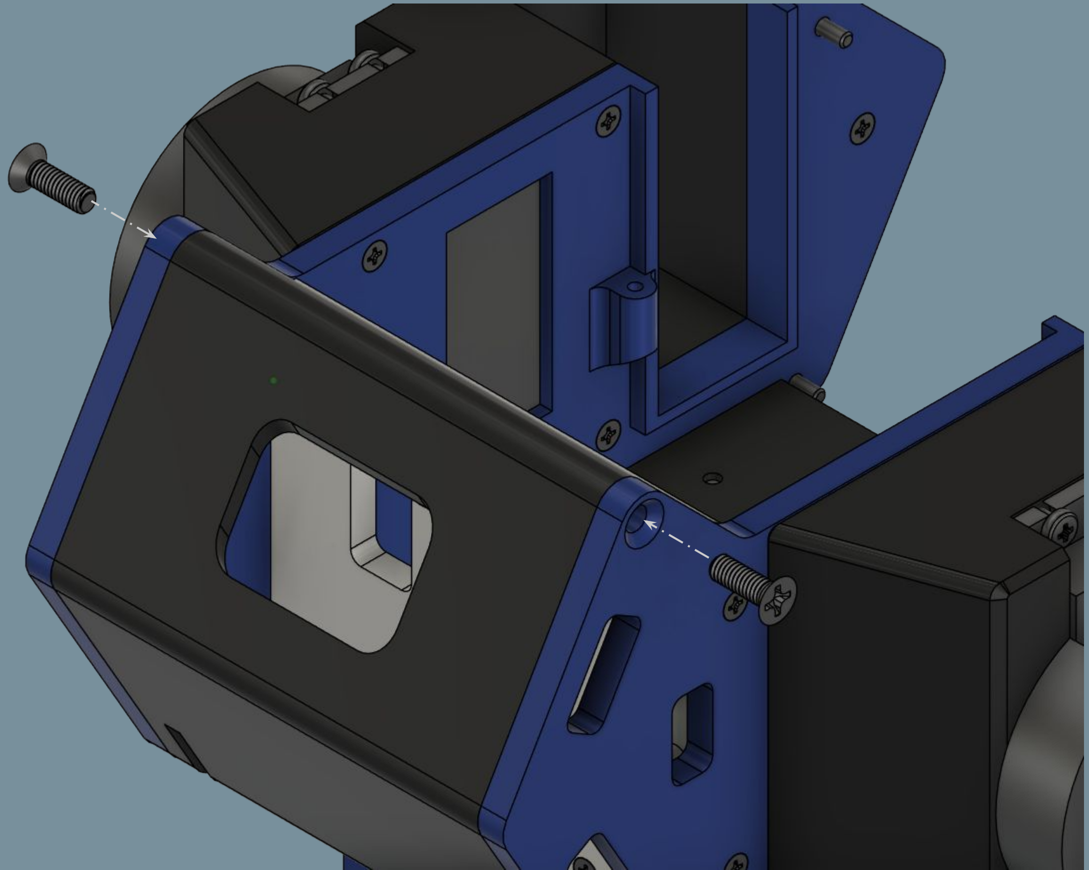
No.	Part Name	Qty
1	TorsoBackPlate	1
2	M2.5x10 Flat Head Screw	2

15c: Side Assembly Mounting

No.	Part Name	Qty
1	M2.5x10 Flat Head Screw	4

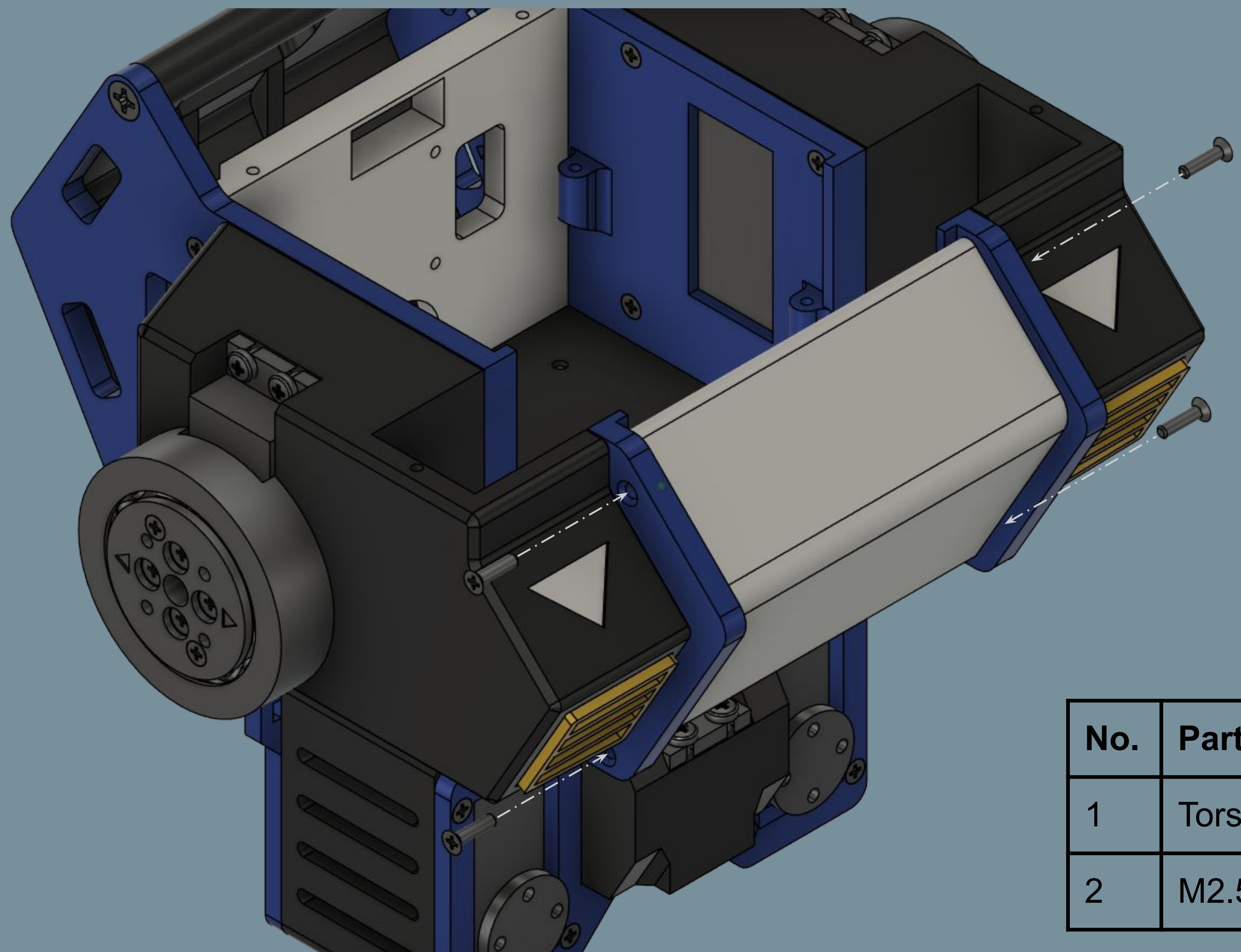


Repeat step for other side



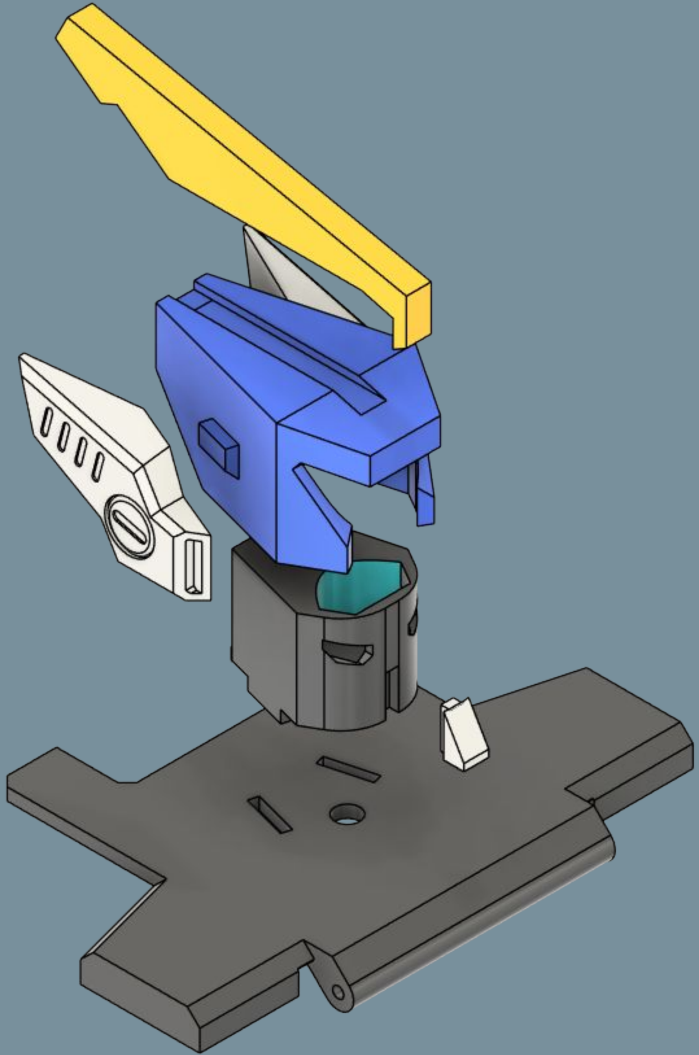
15d: Back Cover Mounting

No.	Part Name	Qty
1	BackCover	1
2	M4x12 Flat Head Screw	2



15e: Front Cover Mounting

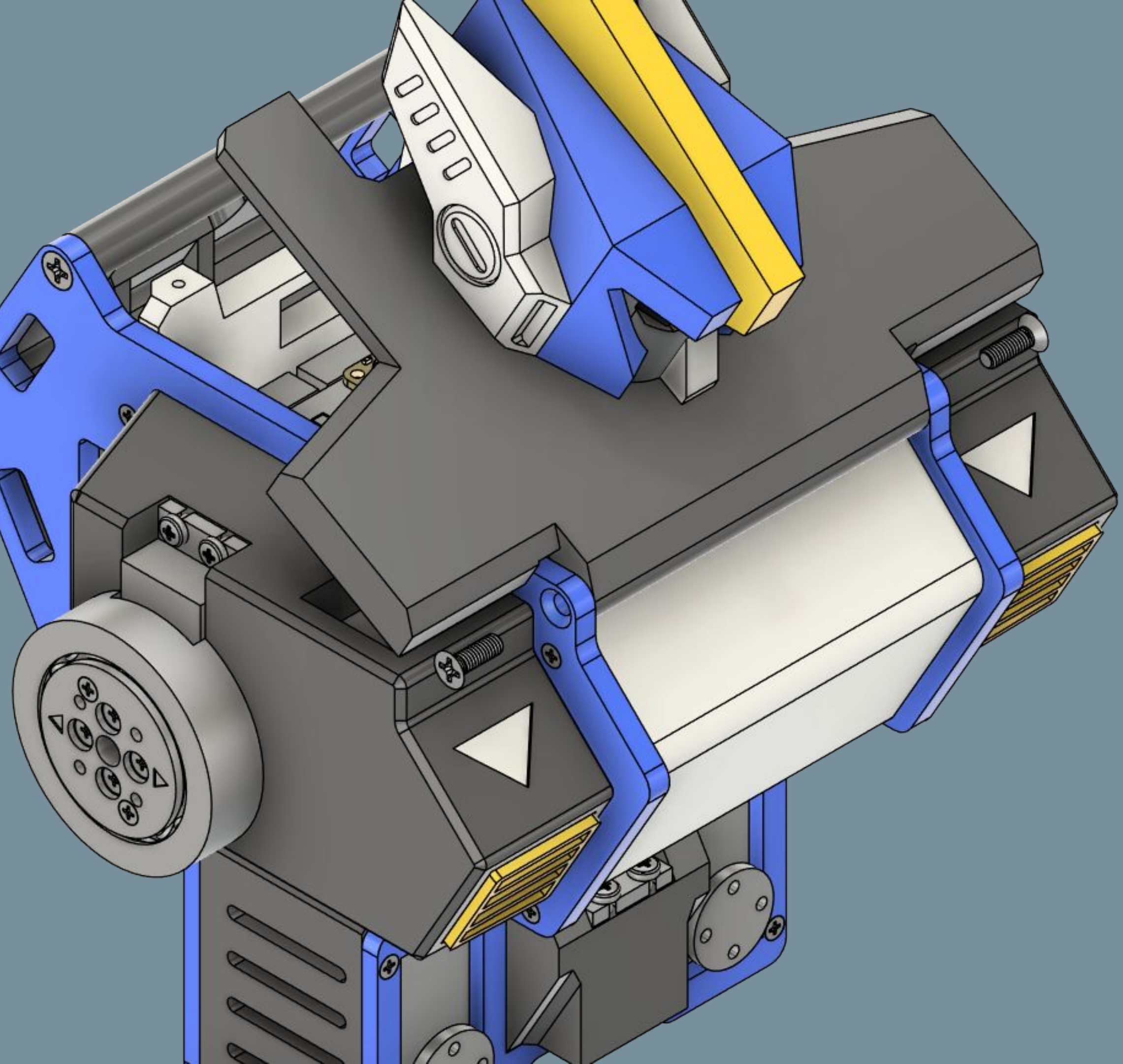
No.	Part Name	Qty
1	TorsoFrontCover	1
2	M2.5x10 Flat Head Screw	4



16a: Top Torso Cover

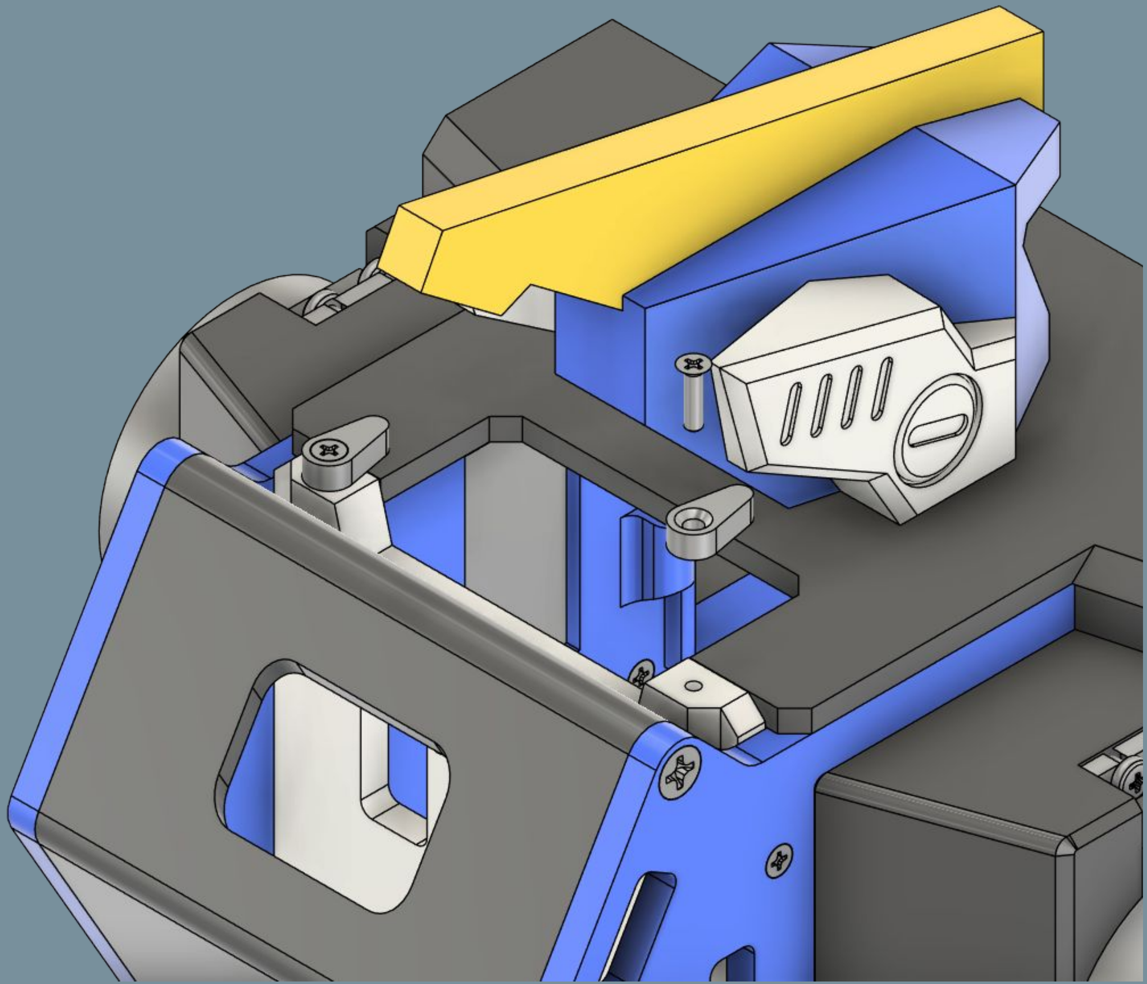
No.	Part Name	Qty
1	TorsoTopCover	1
2	HeadBase	1
3	Helmet	1
4	Chinpiece	1
5	SidepieceL	1
6	SidepieceR	1
7	Crest	1

Use super glue to secure all pieces



16b: Top Torso Cover

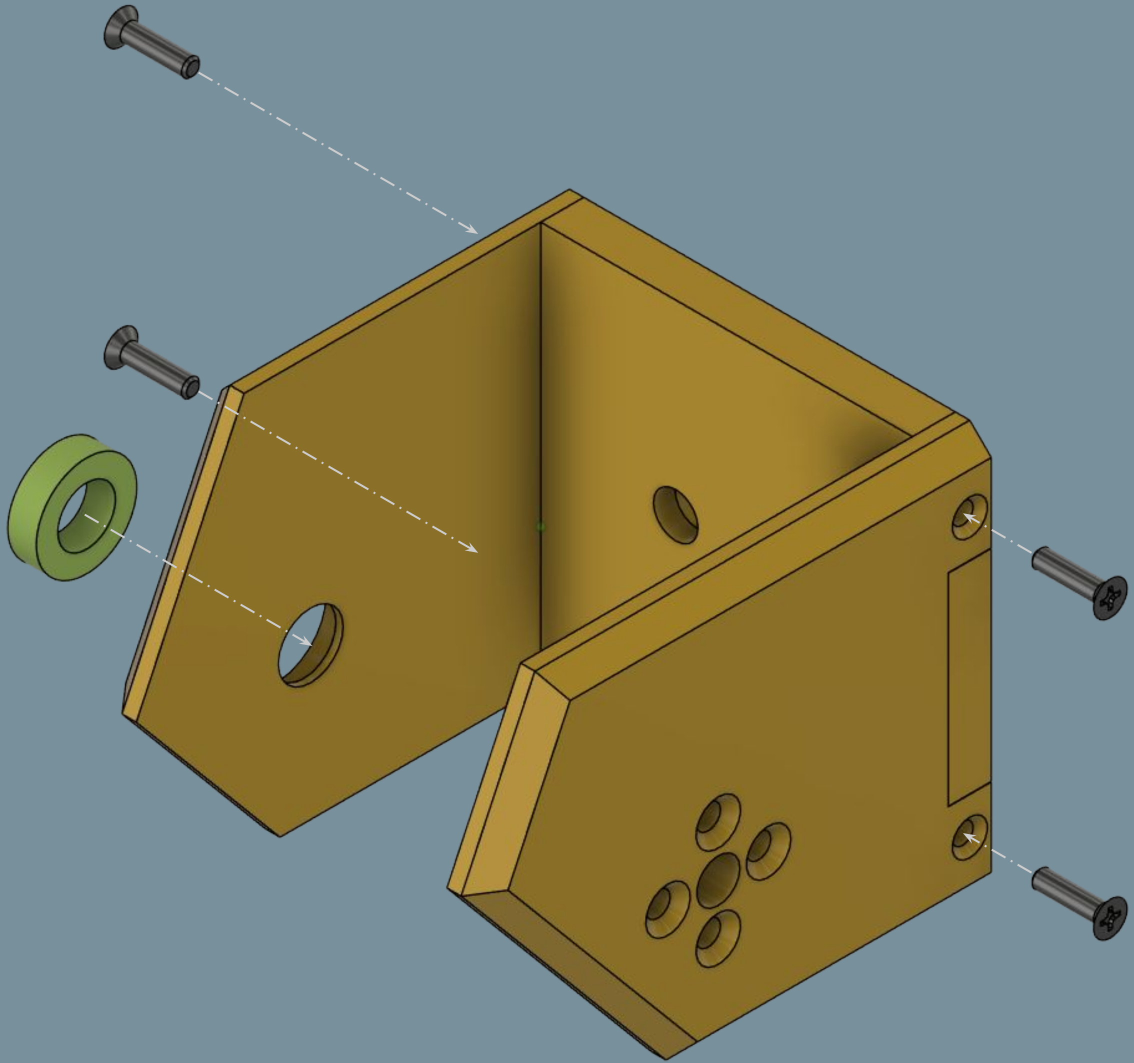
No.	Part Name	Qty
1	M4x12 Flat Head Screw	2



16c: Top Torso Cover

No.	Part Name	Qty
1	LidStopper	2
2	M2.5x10 Flat Head Screw	2

Do not overtighten; lid stoppers must be able to rotate to release the top cover, but not loose.

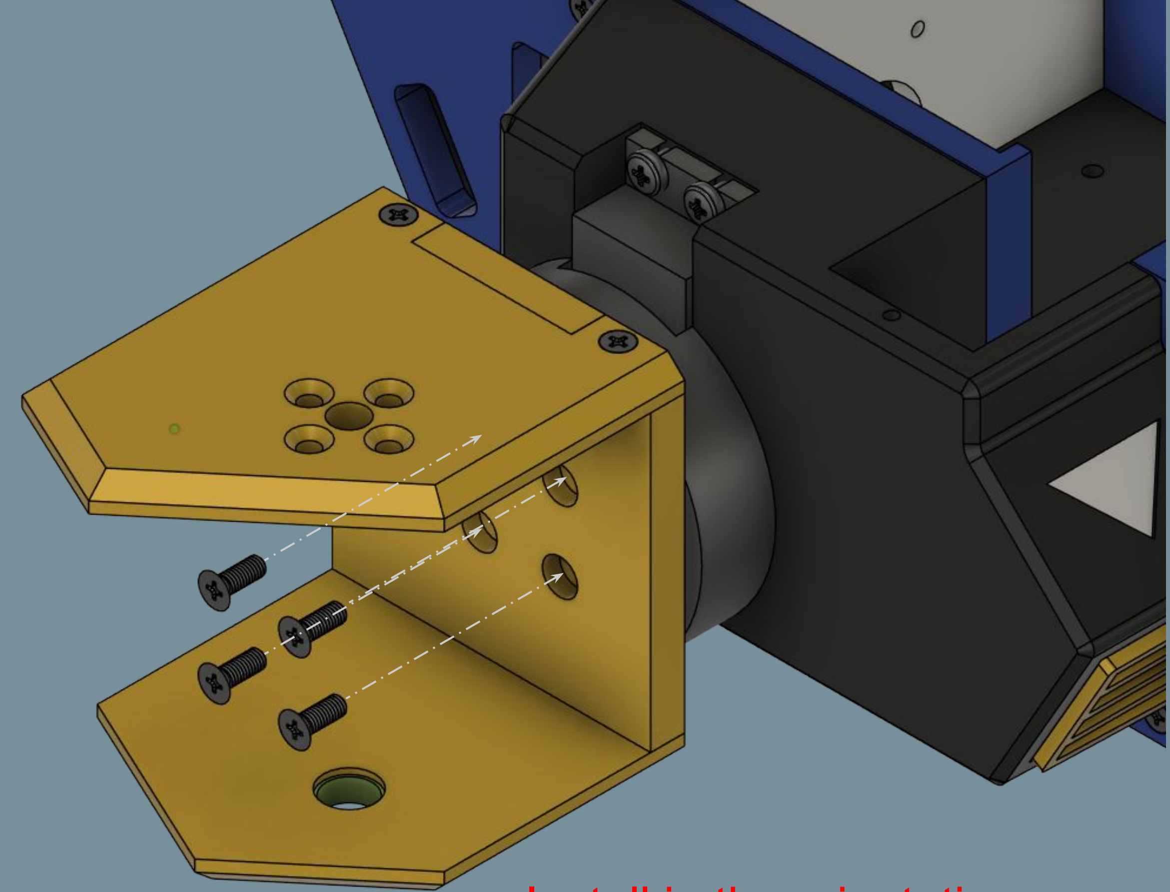


17a: Shoulder Brackets

No.	Part Name	Qty
1	ShoulderBracketBase	1
2	Left/RightShoulderFrontFlange	1
3	Left/RightShoulderRearFlange	1
4	M2.5x10 Flat Head Screw	4
5	8ID 14OD 4T Bearing	1

Right assembly pictured

Repeat step for left assembly



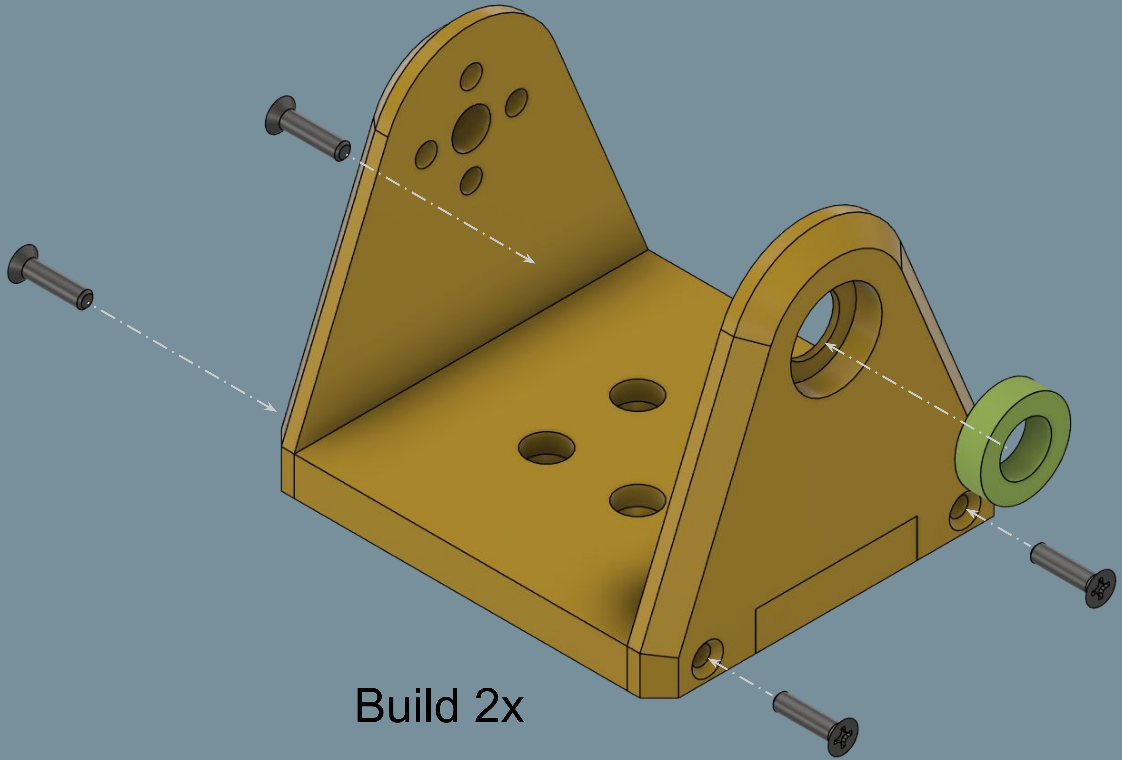
17b: Bracket Mounting

No.	Part Name	Qty
1	M3x8 Flat Head Screw	4

Repeat for other side

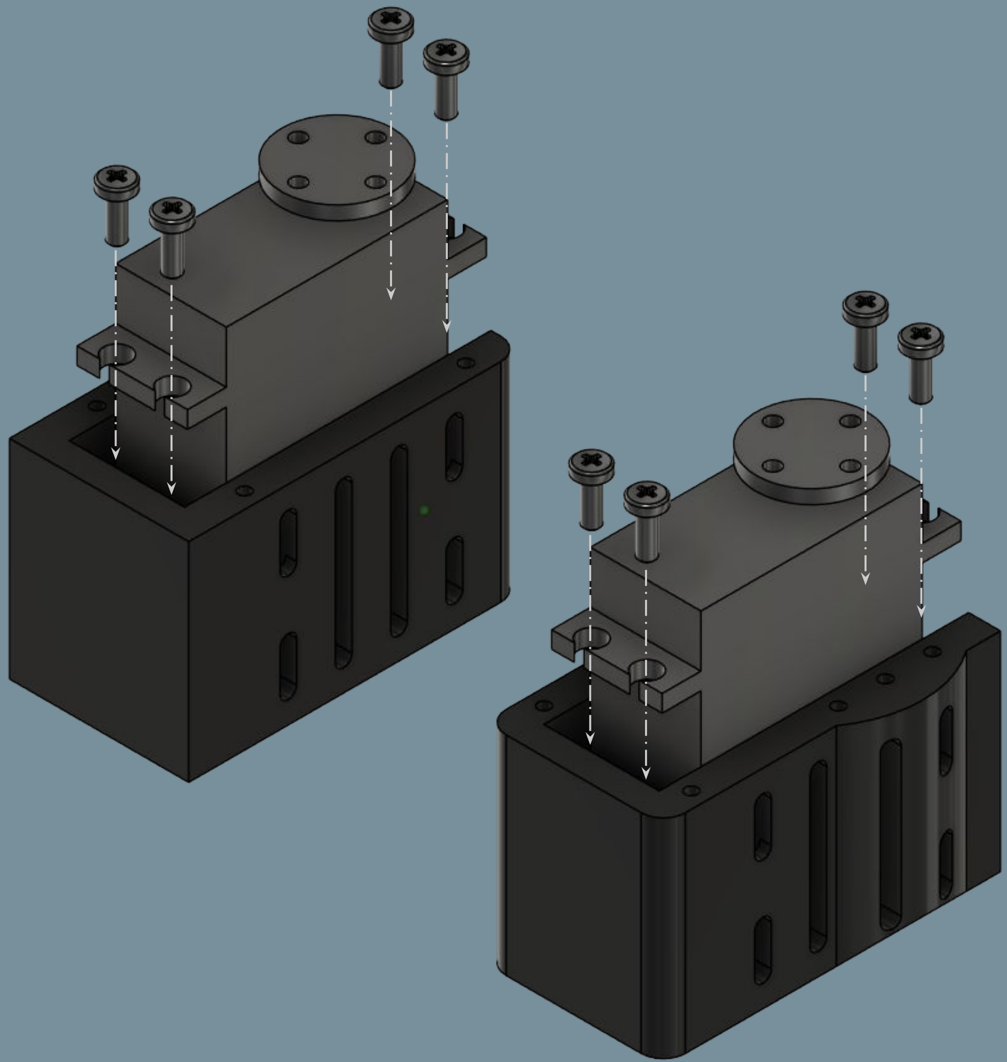
Install in the orientation as shown; this is the servo's zero position

18: Elbow Brackets



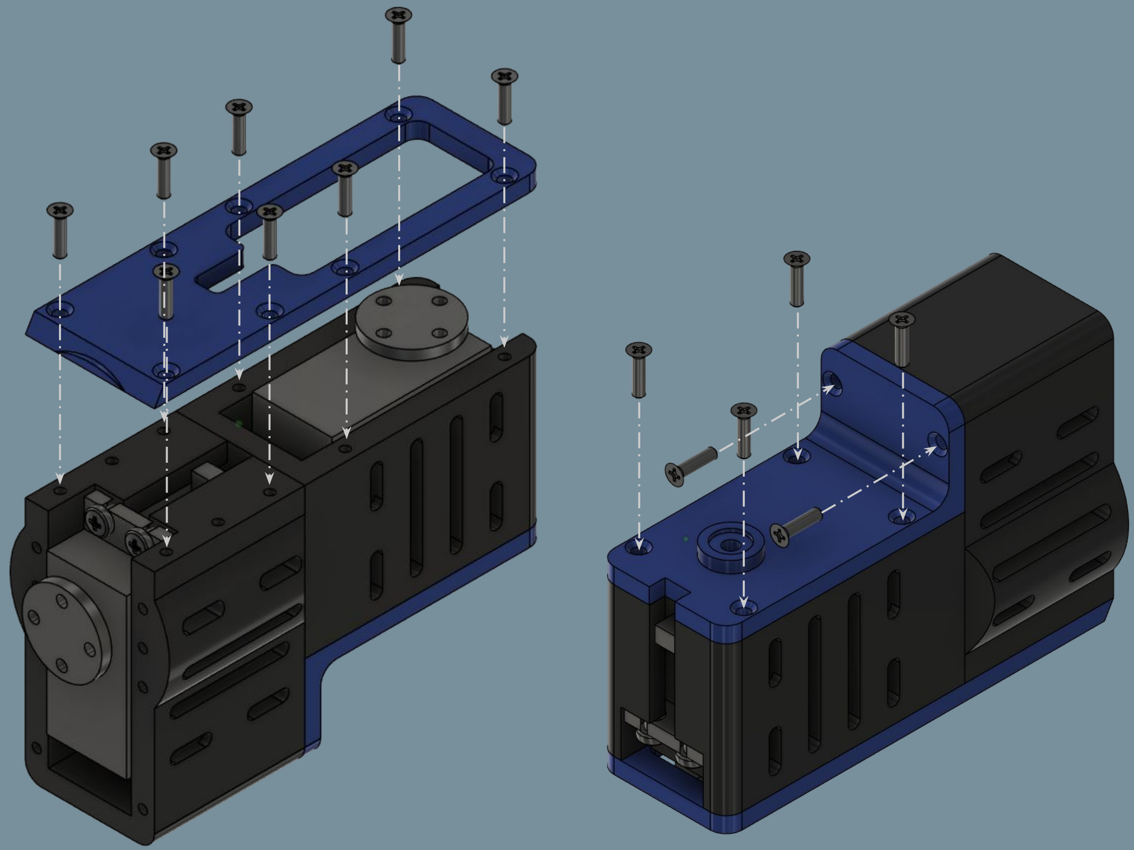
Build 2x

No.	Part Name	Qty
1	ElbowBaseBracket	1(x2)
2	ElbowServoFlange	1(x2)
3	ElbowBearingFlange	1(x2)
4	M2.5x10 Flat Head Screw	4(x2)
5	8ID 14OD 4T Bearing	1(x2)



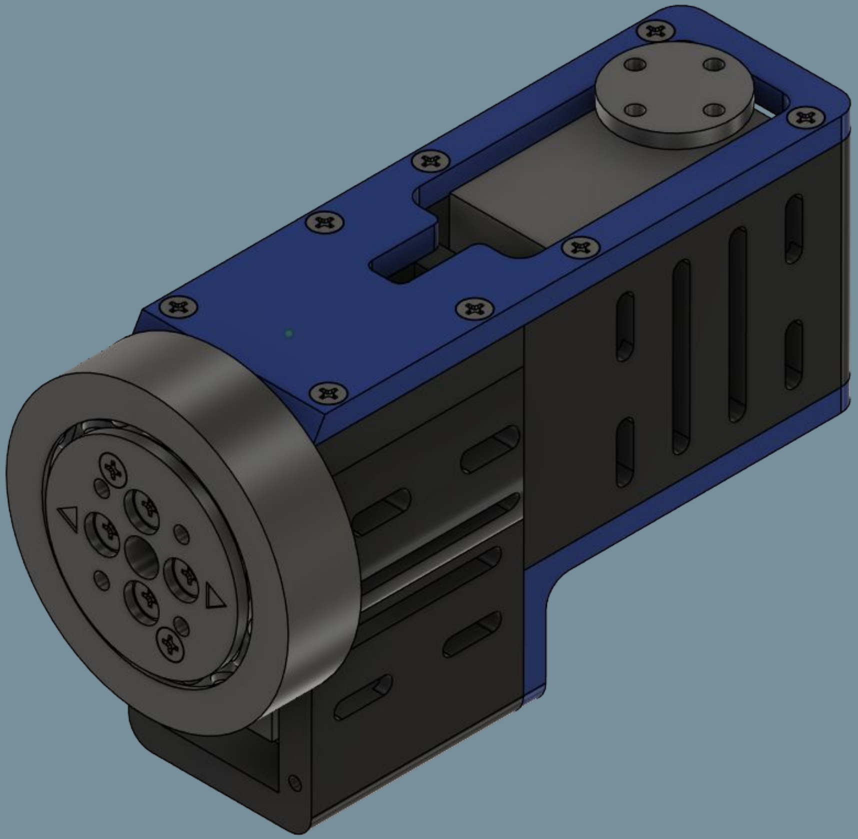
19a: Bicep Servo Housings

No.	Part Name	Qty
1	ShoulderServoHousing	1
2	BicepServoHousing	1
3	DS3235 Servo	2
4	M3x8 Pan Head Screw	8



19b: Bicep Assembly

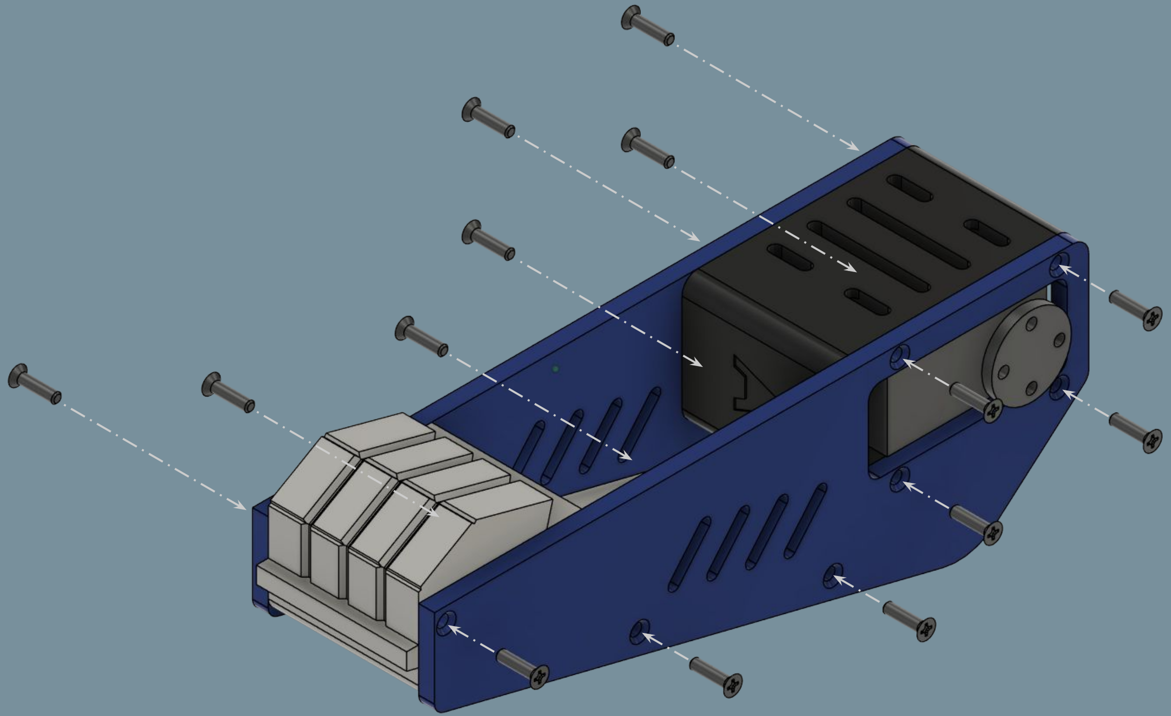
No.	Part Name	Qty
1	BicepPlate	1
2	BicepBracket	1
3	M2.5x10 Flat Head Screw	14



19c: Arm Slew Bearing

Refer to 14b-14d for detailed steps

Repeat steps 19a-19c



20: Forearm Assembly

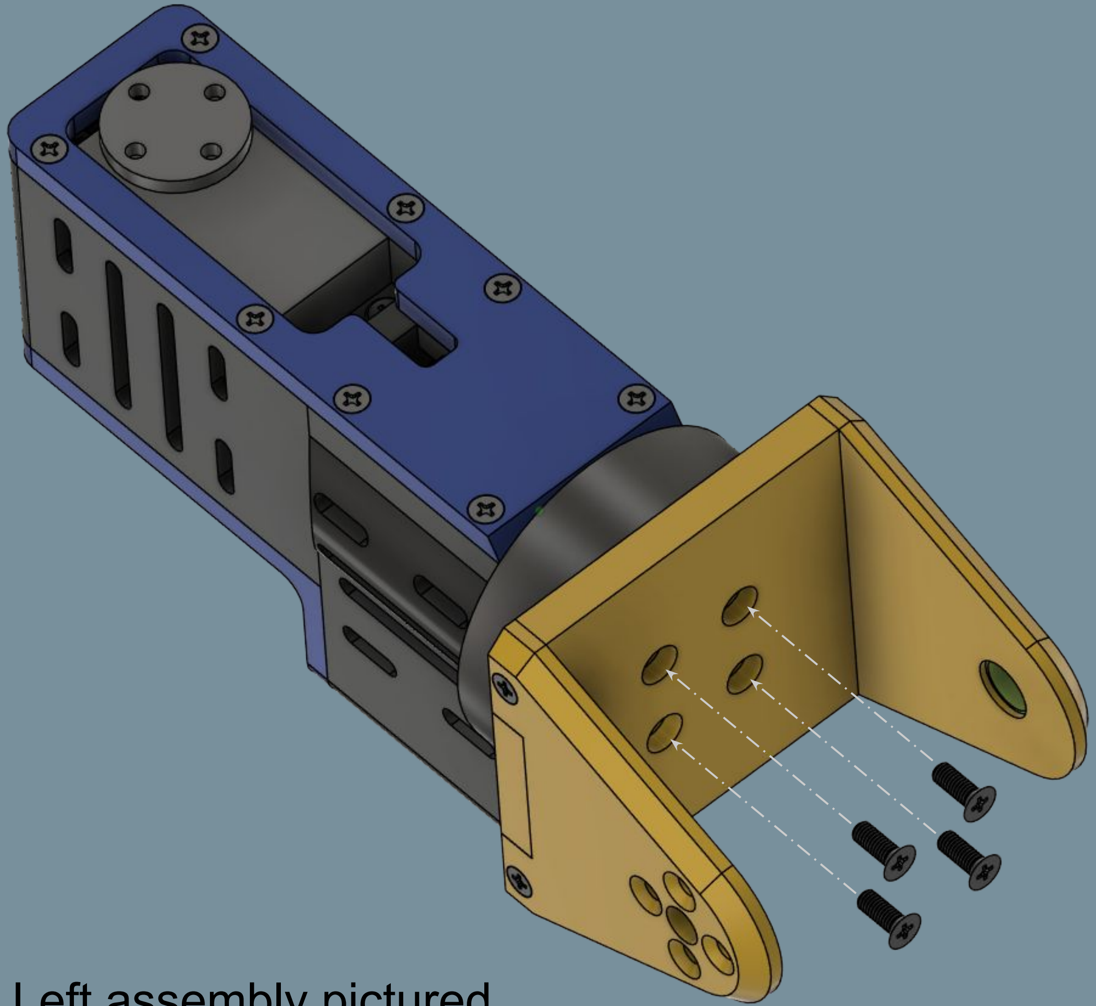
No.	Part Name	Qty
1	Left/RightForearmOuterPlate	1
2	Left/RightForearmInnerPlate	1
3	Fist	1
4	M2.5x10 Flat Head Screw	14

Left assembly pictured

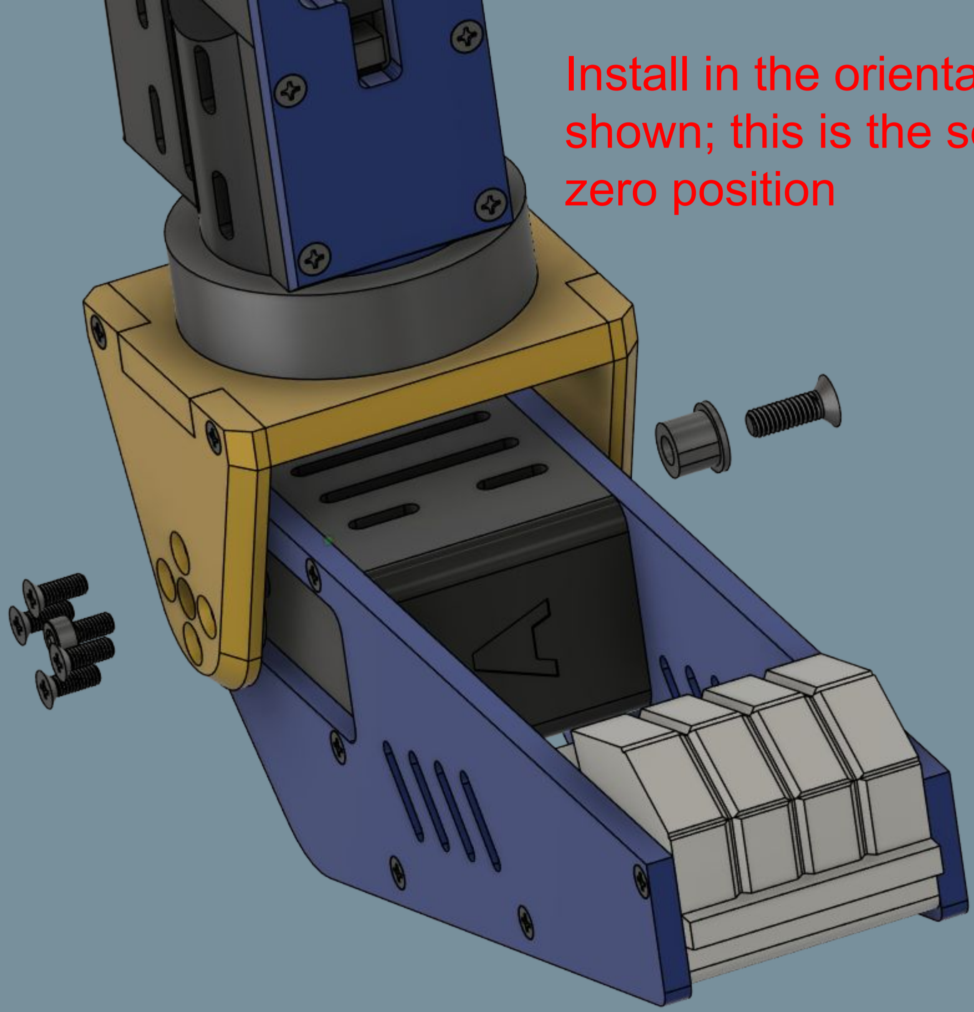
Repeat step for right assembly

21a: Elbow Bracket Mounting

No.	Part Name	Qty
1	M3x8 Flat Head Screw	4



Left assembly pictured

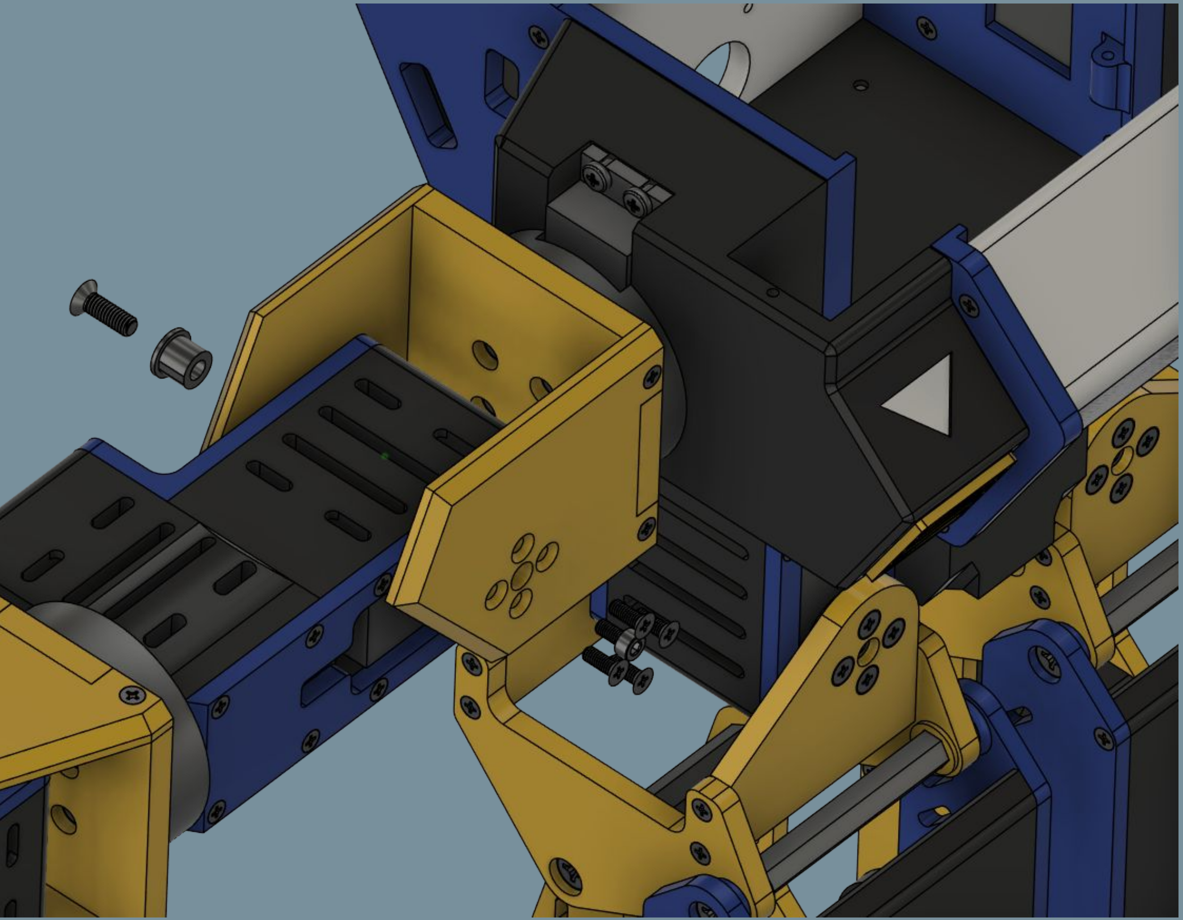


Install in the orientation as shown; this is the servo's zero position

21b: Forearm Mounting

No.	Part Name	Qty
1	M3x6 Socket Head Screw	1
2	M3x8 Flat Head Screw	4
3	M4x12 Flat Head Screw	1
4	ShaftSleeve	1

Left assembly pictured



21c: Arm Mounting

No.	Part Name	Qty
1	M3x6 Socket Head Screw	1
2	M3x8 Flat Head Screw	4
3	M4x12 Flat Head Screw	1
4	ShaftSleeve	1

Install in the orientation as shown; this is the servo's zero position

Repeat steps 21a-21c for other side

Sanity Checklist

- ❏ All joints are structurally stable with minimal wobble and resistance (with exception of motor torque).
- ❏ All joints have been fastened, preferably with reinforcement from a thread bonding agent (e.g. Loctite) to prevent the servo joint from becoming loose.
- ❏ All structural components are fastened properly with no noticeable gaps between component interfaces.